

Supplement to “Surrogate-Assisted Local Conditional Distribution Inference with Scarce Reference Outcomes”

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1 Symbol guide

Symbol	Definition
$O = (W, R, RY)$	Observed data; W includes broadly observed X, S , and $Z_y = 1(Y \leq y)$.
$F_0(y x_0)$	Conditional CDF of the reference outcome at the target point.
$F_{h,0}(y x_0)$	Kernel-regularized target in (1).
$F_{h,\text{emp}}(y x_0)$	Full-data empirical localized CDF used for the EPA and VitalDB controlled-sampling benchmarks.
$\pi_M(W)$	Verification probability; $p_M = E\{\pi_M(W)\}$ and $e_M(W) = \pi_M(W)/p_M$.
$N_{M,h} = Mp_M h^d$	Nominal local-information rate index used in the process normalization.
$q_{0,y}(X), m_{0,y}(W)$	Threshold regressions given X and W , respectively.
$U_y(a)$	Augmented threshold signal in (2).
$\delta_y^a = a_y - m_{0,y}$	Generic augmentation error; $\Delta_{M,y}$ and Δ_y are the triangular and limiting oracle-path errors.
$D_y = m_y - q_y$	Candidate surrogate correction, conditional on the fitting sample.
$A_y = m_{0,y} - q_y$	Correction from the candidate anchor to the true threshold regression.
$J_{M,h}(\lambda)$	Augmentation-dependent integrated local variance along $a_{\lambda,y} = q_y + \lambda D_y$.
$G_{M,h}, H_{M,h}$	Alignment and curvature terms in the quadratic criterion.
λ_h^{or}	Path-oracle coefficient (M implicit); $\hat{\lambda}_{h,k}$ is its fold estimate.
$\hat{\mu}_{h,G,k}, \hat{v}_{h,k}$	Selection-fold kernel mean and empirical transfer weight in (4).
$\mathcal{J}, \mathcal{G}, \mathcal{H}$	Criterion quantities multiplied by p_M for a common fixed/scarce-reference scale.
$\Psi_{M,y}$	Scaled path-oracle influence process in (39).
$\Gamma_p(y, z)$	Covariance kernel of the limiting local CDF process.
$\mathbf{S}_h, L_h$	Local-linear population Gram matrix and equivalent weight.
ϑ	Finite signed measure used for a scalar projection of the threshold-indexed CDF.
$\rho_{M,k}, \rho_k$	Evaluation-fold sample fraction and its positive limit.

Bold symbols denote finite vectors or matrices. A superscript “eff” denotes a canonical-gradient quantity, a superscript “or” denotes a path-oracle estimator with fitted regressions and population borrowing coefficients, and a superscript “cl” denotes an independent-cluster process.

2 Notation and conditions

This section collects the conditions used in the proofs. Exact efficiency and transfer results use weighted square integrability. Gain estimation and path-oracle equivalence add sample separation and local second moments. The continuum process adds threshold increments and entropy control. Point-target and quantile inference add smoothness in x and regularity of the inverse-CDF map.

For each M , let

$$O_{Mi} = (W_i, R_i, R_i Y_i), \quad i = 1, \dots, M,$$

be independent and identically distributed within row M . Here W contains X , S , and any other broadly observed variables. The primary designs use $W = (X, S)$, while an additional observed design variable may enter the verification probability. Write $Z_y = 1(Y \leq y)$,

$$q_{0,y}(X) = E(Z_y | X), \quad m_{0,y}(W) = E(Z_y | W).$$

The joint law of (W, Y) is fixed and the verification law may vary with M . The threshold set $\mathcal{Y} = [\underline{y}, \bar{y}]$ is compact. Define

$$K_h(X - x_0) = K\{(X - x_0)/h\}, \quad \mu_h = EK_h(X - x_0),$$

and

$$F_{h,0}(y | x_0) = \frac{E\{K_h(X - x_0)Z_y\}}{\mu_h}.$$

For an augmentation $a = \{a_y : y \in \mathcal{Y}\}$,

$$U_y(a) = a_y(W) + \frac{R}{\pi_M(W)}\{Z_y - a_y(W)\}.$$

Condition 1 (Sampling and verification). Reference sampling satisfies

$$R \perp Y | W.$$

Define

$$p_M = E\{\pi_M(W)\}, \quad e_M(W) = \pi_M(W)/p_M.$$

Then $p_M \rightarrow p \in [0, 1]$, $E\{e_M(W)\} = 1$, and constants $0 < c_e < C_e < \infty$ satisfy

$$c_e \leq e_M(W) \leq C_e$$

almost surely. The joint local limit of e_M and the process path is specified in Condition 7.

Condition 2 (Local design and kernel). The point x_0 is interior to the support of X . The density f_X is positive and continuous at x_0 . The kernel $K : \mathbb{R}^d \rightarrow [0, \infty)$ is bounded, symmetric, compactly supported, and of bounded variation, with

$$0 < \kappa_1 = \int K(u) du < \infty, \quad 0 < \kappa_2 = \int K^2(u) du < \infty.$$

The bandwidth satisfies

$$h \rightarrow 0, \quad Mp_M h^d \rightarrow \infty.$$

For a threshold-indexed function b , define

$$\|b\|_{\text{ver}}^2 = \sup_{y \in \mathcal{Y}} \frac{p_M h^d}{\mu_h^2} E \left[K_h^2(X - x_0) \left\{ \frac{1}{\pi_M(W)} - 1 \right\} b_y^2(W) \right]. \quad (1)$$

Condition 3 (Candidate paths and sample splitting). The sample is partitioned into three folds I_1, I_2, I_3 , with

$$\rho_{M,j} = |I_j|/M \rightarrow \rho_j \in (0, 1), \quad \rho_1 + \rho_2 + \rho_3 = 1.$$

Regression fitting, weight selection, and CDF evaluation are assigned to different folds and rotate across them. For each evaluation fold k , its threshold regressions and borrowing weight are

constructed from the other two folds. With probability tending to one,

$$\sup_{k,y,x} |\widehat{q}_{k,y}(x)| + \sup_{k,y,w} |\widehat{m}_{k,y}(w)| \leq B$$

for a finite constant B .

Write

$$\varepsilon_M = p_M \varepsilon_{H,M}, \quad \mathcal{H}_{h,k} = p_M H_{M,h,k}.$$

Here and below, dependence on M is implicit for fold-specific population criterion quantities. The theoretical stabilizer satisfies $\varepsilon_{H,M} \downarrow 0$. For every evaluation fold, $\mathcal{H}_{h,k} \rightarrow H_{\infty,k} \in [0, \infty)$. If $H_{\infty,k} > 0$, then

$$\varepsilon_M = o(H_{\infty,k});$$

if $H_{\infty,k} = 0$, the fitted correction path satisfies Condition 6.

Conditional on the fitted regressions, let $\lambda_{h,k}^{\text{or}}$ minimize their population transfer criterion, with dependence on M implicit, and set

$$a_{k,y}^{\text{or}} = \widehat{q}_{k,y} + \lambda_{h,k}^{\text{or}} (\widehat{m}_{k,y} - \widehat{q}_{k,y}).$$

There exist deterministic triangular paths $q_{M,y}(X)$, $m_{M,y}(W)$, their population-minimizing coefficient $\lambda_{M,h}$, and

$$a_{M,y} = q_{M,y} + \lambda_{M,h} (m_{M,y} - q_{M,y})$$

such that, uniformly over k ,

$$\|a_k^{\text{or}} - a_M\|_{\text{ver}} = o_{\mathbb{P}}(1).$$

There is a bounded deterministic path $a_{\infty} = \{a_{\infty,y} : y \in \mathcal{Y}\}$ such that, for every finite C ,

$$\sup_{y \in \mathcal{Y}} \sup_{\|x - x_0\| \leq Ch} E\{(a_{M,y} - a_{\infty,y})^2 \mid X = x\} \rightarrow 0. \quad (2)$$

Write

$$\Delta_{M,y} = a_{M,y} - m_{0,y}, \quad \Delta_y = a_{\infty,y} - m_{0,y}.$$

Condition 4 (Local regression stability). For each evaluation fold k , conditional on the fitting data,

$$\|\widehat{q}_k - q_M\|_{\text{ver}} = o_{\mathbb{P}}(1), \quad (3)$$

$$\|\widehat{m}_k - m_M\|_{\text{ver}} = o_{\mathbb{P}}(1). \quad (4)$$

The realized difference classes are pointwise measurable and satisfy Condition 5, with envelopes whose local L_2 norms are $o_{\mathbb{P}}(1)$.

Condition 5 (Threshold increments and entropy). There exist constants $C_{\mathcal{Y}} < \infty$ and $\alpha > 0$ such that, locally around x_0 ,

$$\Pr(y < Y \leq z \mid W) \leq C_{\mathcal{Y}}|z - y| \quad (5)$$

for $y < z$, and

$$E\{(a_{M,z} - a_{M,y})^2 \mid X = x\} \leq C_{\mathcal{Y}}|z - y|^{2\alpha}. \quad (6)$$

The path-oracle influence classes and fitted difference classes are VC-type uniformly in M : for every finitely discrete probability measure Q ,

$$N\{\epsilon\|F_M\|_{Q,2}, \mathcal{F}_M, L_2(Q)\} \leq (A/\epsilon)^V, \quad 0 < \epsilon \leq 1, \quad (7)$$

for constants $A, V < \infty$ independent of M . For the path-oracle envelope,

$$\sup_M P F_M^2 < \infty$$

and, for every $\eta > 0$,

$$P\{F_M^2 1(F_M > \eta\sqrt{M})\} \rightarrow 0. \quad (8)$$

The fitted difference classes satisfy the conditional analogues with local L_2 envelopes of order $o_{\mathbb{P}}(1)$.

Condition 6 (Path-norm compatibility). Let $D_{M,y} = m_{M,y} - q_{M,y}$, and define

$$\mathcal{H}_{M,h} = p_M E\{v_{M,h} \|D_M\|_v^2\},$$

$$\mathcal{D}_{M,h}^2 = \sup_{y \in \mathcal{Y}} \frac{p_M h^d}{\mu_h^2} E \left[K_h^2 \left(\frac{1}{\pi_M} - 1 \right) D_{M,y}^2 \right].$$

There is a deterministic nondecreasing function ω , continuous at zero with $\omega(0) = 0$, such that

$$\mathcal{D}_{M,h}^2 \leq \omega(\mathcal{H}_{M,h}) + o(1).$$

The same relation holds in probability for each fitted correction path on its gain and evaluation folds.

Condition 7 (Covariance convergence). Uniformly over $(y, z) \in \mathcal{Y}^2$, define

$$B_{M,yz}(W) = \frac{C_{yz}(W)}{e_M(W)} + \left\{ \frac{1}{e_M(W)} - p_M \right\} \Delta_{M,y}(W) \Delta_{M,z}(W)$$

$$+ p_M \{m_{0,y}(W) - F_{h,0}(y | x_0)\} \{m_{0,z}(W) - F_{h,0}(z | x_0)\},$$

$$B_{0,yz}(W) = \frac{C_{yz}(W)}{e_0(W)} + \left\{ \frac{1}{e_0(W)} - p \right\} \Delta_y(W) \Delta_z(W)$$

$$+ p \{m_{0,y}(W) - F_0(y | x_0)\} \{m_{0,z}(W) - F_0(z | x_0)\}.$$

For every finite C ,

$$\sup_{\|x-x_0\| \leq Ch} \sup_{y,z \in \mathcal{Y}} |E\{B_{M,yz}(W) | X = x\} - E\{B_{0,yz}(W) | X = x_0\}| \rightarrow 0. \quad (9)$$

Set

$$\Gamma_p(y, z) = \frac{\kappa_K}{f_X(x_0)} E\{B_{0,yz}(W) | X = x_0\}. \quad (10)$$

The kernel Γ_p is continuous and defines a tight Gaussian process. For supremum inference, the distribution of $\|\mathbb{G}_p\|_{\mathcal{Y}}$ is continuous at the target critical value. Finite-dimensional studentization is applied only to projections with positive limiting variance.

Condition 8 (Localization bias and quantiles). For inference on F_0 , the maps $x \mapsto q_{0,y}(x)$ and $x \mapsto f_X(x)$ have bounded continuous second derivatives near x_0 , uniformly over $y \in \mathcal{Y}$. For quantile inference, $F_0(\cdot \mid x_0)$ is continuously differentiable and strictly increasing on $\mathcal{Y}$, its range is contained in $[\eta, 1 - \eta]$ for some $\eta > 0$, and

$$0 < c_f \leq f_{Y|X}(y \mid x_0) \leq C_f < \infty, \quad y \in \mathcal{Y}.$$

Main-article assumption	Detailed Supplement conditions
Assumption 1: observed-data identification and verification	Triangular-array setup and 1
Assumption 2: localization and the local-information rate	2
Assumption 3: candidate paths and sample separation	3, 4, (1), and (2)
Assumption 4: transfer-path curvature and degeneracy	Curvature clauses in 3 and 6
Assumption 5: threshold-process regularity	5 and 7
Assumption 6: point-target and quantile regularity	8

Main result	Conditions used
Canonical gradient and efficiency	1–2
Transfer criterion	Weighted square integrability of $J_{M,h}$
Adaptive path-oracle equivalence	1–5 and 6
Uniform CDF process	1–7
Multiplier process	1–7
Point target and quantiles	1–8

3 Primitive conditions for covariance-kernel convergence

Condition 7 is stated directly in terms of the covariance integrand used by the local process. The following proposition gives lower-level sufficient conditions for its kernel-convergence and continuity components.

Proposition 1 (Primitive covariance-kernel convergence). *Suppose Conditions 1–3 hold. Assume, for every finite C ,*

$$\sup_{\|x-x_0\| \leq Ch} E \left[\left| e_M^{-1}(W) - e_0^{-1}(W) \right|^2 \mid X = x \right] \rightarrow 0, \quad (11)$$

$$\sup_{y \in \mathcal{Y}} \sup_{\|x-x_0\| \leq Ch} E \left[\{a_{M,y}(W) - a_{\infty,y}(W)\}^2 \mid X = x \right] \rightarrow 0, \quad (12)$$

$$\sup_{y \in \mathcal{Y}} |F_{h,0}(y \mid x_0) - F_0(y \mid x_0)| \rightarrow 0. \quad (13)$$

Assume the following conditional moment maps are uniformly continuous at x_0 , uniformly over $(y, z) \in \mathcal{Y}^2$:

$$\begin{aligned} x &\longmapsto E \left\{ \frac{C_{yz}(W)}{e_0(W)} \mid X = x \right\}, \\ x &\longmapsto E \left[\left\{ \frac{1}{e_0(W)} - p \right\} \Delta_y(W) \Delta_z(W) \mid X = x \right], \\ x &\longmapsto E \left[\{m_{0,y}(W) - F_0(y \mid x_0)\} \{m_{0,z}(W) - F_0(z \mid x_0)\} \mid X = x \right]. \end{aligned}$$

At $x = x_0$, these three conditional moment functions are continuous in (y, z) . If a_M and a_∞ are uniformly bounded, then the covariance-kernel convergence and continuity requirements in Condition 7 hold. Tight Gaussian realizability and continuity of the supremum law remain the process-level requirements stated separately in that condition.

Proof. Write $B_{M,yz} = T_{1,M,yz} + T_{2,M,yz} + T_{3,M,yz}$ for the three terms in Condition 7. Since $|C_{yz}(W)| \leq 1$

and e_M, e_0 are bounded away from zero, conditional Cauchy–Schwarz and (11) give

$$\sup_{\|x-x_0\|\leq Ch} \sup_{y,z} \left| E(T_{1,M,yz} \mid X=x) - E\left\{ \frac{C_{yz}}{e_0} \mid X=x \right\} \right| \rightarrow 0.$$

Uniform continuity of the limiting conditional moment then replaces x by x_0 .

For the augmentation term, boundedness and

$$\Delta_{M,y} - \Delta_y = a_{M,y} - a_{\infty,y}$$

imply, uniformly over y, z ,

$$|\Delta_{M,y}\Delta_{M,z} - \Delta_y\Delta_z| \leq C\{|a_{M,y} - a_{\infty,y}| + |a_{M,z} - a_{\infty,z}|\}.$$

Equations (11)–(12), conditional Cauchy–Schwarz, and $p_M \rightarrow p$ therefore give

$$\sup_{\|x-x_0\|\leq Ch} \sup_{y,z} \left| E(T_{2,M,yz} \mid X=x) - E\left[\left\{ \frac{1}{e_0} - p \right\} \Delta_y \Delta_z \mid X=x \right] \right| \rightarrow 0.$$

Uniform continuity again replaces x by x_0 .

Finally, $m_{0,y} \in [0, 1]$, $p_M \rightarrow p$, and (13) imply

$$\sup_{\|x-x_0\|\leq Ch} \sup_{y,z} \left| E(T_{3,M,yz} \mid X=x) - pE\left[\{m_{0,y} - F_0(y \mid x_0)\}\{m_{0,z} - F_0(z \mid x_0)\} \mid X=x \right] \right| \rightarrow 0.$$

The stated uniform continuity gives the target-stratum limit. Summing the three bounds proves (9); continuity in (y, z) gives continuity of Γ_p . These are exactly the covariance-kernel components claimed in the proposition. $\square$

4 Efficiency bound and augmentation covariance

Fix M and h . The observed-data law factors as

$$p_O(o) = p_W(w) \{ \pi_M(w) p_{Y|W}(y | w) \}^r \{ 1 - \pi_M(w) \}^{1-r},$$

where the factor involving Y is present only when $r = 1$. With known π_M , the nonparametric tangent space is the $L_2(P)$ -closure of

$$s(O) = s_W(W) + R s_Y(Y, W), \quad E s_W(W) = 0, \quad E \{ s_Y(Y, W) | W \} = 0. \quad (14)$$

When the verification law is unrestricted, the tangent space additionally contains

$$s_\pi(O) = \{ R - \pi_M(W) \} c(W) \quad (15)$$

for square-integrable c . This is the observed-data tangent decomposition for missing-at-random models; see Robins et al. (1994), Chapters 3–4 of Tsiatis (2006), and Chapter 25 of van der Vaart (1998).

For a finite signed measure ϑ on $\mathcal{Y}$, let

$$\zeta_\vartheta(Y) = \int_{\mathcal{Y}} 1(Y \leq y) d\vartheta(y), \quad m_{0,\vartheta}(W) = E \{ \zeta_\vartheta(Y) | W \},$$

and

$$\theta_{M,h,\vartheta} = \frac{E \{ K_h(X - x_0) \zeta_\vartheta(Y) \}}{\mu_h}.$$

Finite total variation of ϑ makes ζ_ϑ bounded.

Theorem 1 (Signed-projection canonical gradient). *Under Condition 1, assume $\mu_h > 0$ and*

$E\{K_h^2(X - x_0)/\pi_M(W)\} < \infty$. The canonical gradient for $\theta_{M,h,\vartheta}$ is

$$\phi_{M,h,\vartheta}^{\text{eff}}(O) = \frac{K_h(X - x_0)}{\mu_h} \left[m_{0,\vartheta}(W) + \frac{R}{\pi_M(W)} \{\zeta_\vartheta(Y) - m_{0,\vartheta}(W)\} - \theta_{M,h,\vartheta} \right]. \quad (16)$$

The gradient is orthogonal to the propensity-score tangent space (15). For a finite threshold collection $y_1, \dots, y_L$ and $\vartheta = \sum_{j=1}^L c_j \delta_{y_j}$,

$$\phi_{M,h,\vartheta}^{\text{eff}} = \sum_{j=1}^L c_j \phi_{M,h,y_j}^{\text{eff}}.$$

Proof. Consider a regular parametric submodel with score $s = s_W + R s_Y$ as in (14). Write

$$\mathcal{N}_{M,h,\vartheta} = E\{K_h(X - x_0)\zeta_\vartheta(Y)\}, \quad \theta_{M,h,\vartheta} = \mathcal{N}_{M,h,\vartheta}/\mu_h.$$

Differentiating the numerator and denominator gives

$$\dot{\mathcal{N}}_{M,h,\vartheta} = E\{K_h m_{0,\vartheta} s_W\} + E(K_h \zeta_\vartheta s_Y), \quad (17)$$

$$\dot{\mu}_h = E(K_h s_W). \quad (18)$$

Hence

$$\dot{\theta}_{M,h,\vartheta} = \frac{1}{\mu_h} \left[E\{K_h(m_{0,\vartheta} - \theta_{M,h,\vartheta})s_W\} + E(K_h \zeta_\vartheta s_Y) \right]. \quad (19)$$

The function in (16) has mean zero and lies in the closure of the tangent space. Its W -component is

$$\frac{K_h}{\mu_h}(m_{0,\vartheta} - \theta_{M,h,\vartheta}),$$

and its conditional-outcome component is

$$\frac{K_h}{\mu_h \pi_M}(\zeta_\vartheta - m_{0,\vartheta}).$$

For the W -score,

$$E(\phi_{M,h,\vartheta}^{\text{eff}} s_W) = \frac{1}{\mu_h} E\{K_h(m_{0,\vartheta} - \theta_{M,h,\vartheta}) s_W\}.$$

Using $R \perp Y \mid W$, $E(s_Y \mid W) = 0$, and $R^2 = R$,

$$\begin{aligned} E(\phi_{M,h,\vartheta}^{\text{eff}} R s_Y) &= \frac{1}{\mu_h} E \left[K_h \frac{R}{\pi_M} (\zeta_{\vartheta} - m_{0,\vartheta}) s_Y \right] \\ &= \frac{1}{\mu_h} E(K_h \zeta_{\vartheta} s_Y). \end{aligned}$$

The sum equals (19), so $\phi_{M,h,\vartheta}^{\text{eff}}$ is the canonical gradient.

For a propensity score s_{π} in (15),

$$\begin{aligned} E\{\phi_{M,h,\vartheta}^{\text{eff}} s_{\pi} \mid W\} &= \frac{K_h c(W)}{\mu_h} E \left[\frac{R}{\pi_M} (\zeta_{\vartheta} - m_{0,\vartheta}) (R - \pi_M) \mid W \right] \\ &= \frac{K_h c(W)}{\mu_h} \{1 - \pi_M(W)\} E(\zeta_{\vartheta} - m_{0,\vartheta} \mid W) = 0. \end{aligned}$$

Thus the canonical gradient is unchanged when π_M is included in the nuisance model. Linearity gives the finite-threshold vector result. $\square$

Theorem 4 follows by taking ϑ to be the coordinate point masses $\delta_{y_1}, \dots, \delta_{y_L}$.

Lemma 1 (Augmentation identity). *Under missing at random, for every integrable augmentation $a_y(W)$,*

$$E\{U_y(a) \mid W\} = m_{0,y}(W), \quad E\{U_y(a) \mid X\} = q_{0,y}(X).$$

Proof. Conditional on W ,

$$E(RZ_y \mid W) = \pi_M(W) m_{0,y}(W), \quad E(R \mid W) = \pi_M(W).$$

Therefore

$$\begin{aligned} E\{U_y(a) \mid W\} &= a_y(W) + \frac{1}{\pi_M(W)} E[R\{Z_y - a_y(W)\} \mid W] \\ &= a_y(W) + m_{0,y}(W) - a_y(W) = m_{0,y}(W). \end{aligned}$$

Taking conditional expectation given X yields the second identity. $\square$

Lemma 2 (Conditional covariance decomposition). *Let $\delta_y^a(W) = a_y(W) - m_{0,y}(W)$, and assume the displayed conditional second moments are finite. Then*

$$\text{Cov}\{U_y(a), U_z(a) \mid W\} = \frac{C_{yz}(W)}{\pi_M(W)} + \left\{ \frac{1}{\pi_M(W)} - 1 \right\} \delta_y^a(W) \delta_z^a(W). \quad (20)$$

Proof. Write

$$U_y(a) - m_{0,y} = \frac{R}{\pi_M} (Z_y - m_{0,y}) + \delta_y^a \left(1 - \frac{R}{\pi_M} \right). \quad (21)$$

The two terms have conditional mean zero. Their cross moment vanishes because $R \perp Y \mid W$ and $E(Z_y - m_{0,y} \mid W) = 0$. Since $R^2 = R$,

$$E \left[\frac{R^2}{\pi_M^2} (Z_y - m_{0,y})(Z_z - m_{0,z}) \mid W \right] = \frac{C_{yz}(W)}{\pi_M(W)}.$$

Also,

$$E \left[\left(1 - \frac{R}{\pi_M} \right)^2 \mid W \right] = \frac{1}{\pi_M(W)} - 1.$$

Substitution into (21) proves (20). $\square$

Proof of Corollary 2. For a fixed augmentation a , define

$$\phi_{M,h,y}^a(O) = \frac{K_h}{\mu_h} \{U_y(a) - F_{h,0}(y \mid x_0)\}.$$

The difference from the canonical gradient is

$$\Delta_{M,h,y}^a(O) = \phi_{M,h,y}^a(O) - \phi_{M,h,y}^{m_0}(O) = \frac{K_h}{\mu_h} \delta_y^a(W) \left(1 - \frac{R}{\pi_M(W)} \right). \quad (22)$$

For known π_M , $\Delta_{M,h,y}^a$ is orthogonal to the tangent space in (14). For every s_W ,

$$E(\Delta_{M,h,y}^a s_W \mid W) = \frac{K_h}{\mu_h} \delta_y^a s_W E \left(1 - \frac{R}{\pi_M} \mid W \right) = 0.$$

For every s_Y with $E(s_Y | W) = 0$,

$$E(\Delta_{M,h,y}^a R s_Y | W) = 0.$$

Hence $\phi_{M,h,y}^a$ is an influence function in the known-design model and $\Delta_{M,h,y}^a$ is its orthogonal excess component.

When π_M is unrestricted,

$$E\{\Delta_{M,h,y}^a s_\pi | W\} = -\frac{K_h}{\mu_h} \delta_y^a(W) c(W) \{1 - \pi_M(W)\}. \quad (23)$$

This inner product vanishes for every c exactly when

$$K_h(X - x_0) \{1 - \pi_M(W)\} \delta_y^a(W) = 0 \quad \text{a.s.}$$

The oracle augmentation therefore retains the canonical-gradient property under both known and unknown verification laws, while the family ϕ^a is the influence-function family for the known-design experiment.

By Lemma 2,

$$\Omega_{M,h}^a(y, z) - \Omega_{M,h}^{m_0}(y, z) = \frac{h^d}{\mu_h^2} E \left[K_h^2 \left(\frac{1}{\pi_M} - 1 \right) \delta_y^a \delta_z^a \right].$$

For thresholds $y_1, \dots, y_L$ and $c \in \mathbb{R}^L$,

$$\begin{aligned} & \sum_{j=1}^L \sum_{\ell=1}^L c_j c_\ell \{ \Omega_{M,h}^a(y_j, y_\ell) - \Omega_{M,h}^{m_0}(y_j, y_\ell) \} \\ &= \frac{h^d}{\mu_h^2} E \left[K_h^2 \left(\frac{1}{\pi_M} - 1 \right) \left\{ \sum_{j=1}^L c_j \delta_{y_j}^a \right\}^2 \right] \geq 0. \end{aligned}$$

This proves the positive-semidefinite covariance decomposition. □

Proof of Theorem 5. Along $a_{\lambda,y} = q_y + \lambda D_y$,

$$a_{\lambda,y} - m_{0,y} = -(A_y - \lambda D_y).$$

Consequently,

$$\begin{aligned} J_{M,h}(\lambda) &= E\{v_{M,h}\|A - \lambda D\|_v^2\} \\ &= E(v_{M,h}\|A\|_v^2) - 2\lambda E\{v_{M,h}\langle A, D \rangle_v\} + \lambda^2 E(v_{M,h}\|D\|_v^2) \\ &= J_{M,h}(0) - 2\lambda G_{M,h} + \lambda^2 H_{M,h}. \end{aligned}$$

If $H_{M,h} = 0$, weighted Cauchy–Schwarz gives

$$|G_{M,h}|^2 \leq J_{M,h}(0)H_{M,h} = 0.$$

Thus $G_{M,h} = 0$, the criterion is constant on $[0, 1]$, and the convention $\lambda_h^{\text{or}} = 0$ applies.

If $H_{M,h} > 0$, the unconstrained minimizer is $G_{M,h}/H_{M,h}$, and projection gives λ_h^{or} . Setting $\lambda = 1$ gives

$$J_{M,h}(0) - J_{M,h}(1) = 2G_{M,h} - H_{M,h}.$$

The remaining cases in (15) follow according to whether the projected minimizer equals zero, lies in $(0, 1)$, or equals one. □

5 Estimation of the borrowing criterion

Lemma 3 (Kernel moments and denominator). *Under Condition 2,*

$$\mu_h = h^d f_X(x_0) \kappa_1 \{1 + o(1)\}, \tag{24}$$

$$EK_h^2(X - x_0) = h^d f_X(x_0) \kappa_2 \{1 + o(1)\}. \tag{25}$$

If I is a fold of size n_I with $n_I/M \rightarrow \rho_I \in (0, 1)$, then

$$\frac{\mathbb{P}_I K_h}{\mu_h} - 1 = O_{\mathbb{P}}\{(n_I h^d)^{-1/2}\} = O_{\mathbb{P}}\{(M h^d)^{-1/2}\}. \quad (26)$$

Proof. Changing variables $x = x_0 + hu$,

$$\mu_h = h^d \int K(u) f_X(x_0 + hu) du.$$

Continuity of f_X , compact support of K , and dominated convergence give (24); the same argument gives (25). Moreover,

$$\text{Var}(\mathbb{P}_I K_h) = \frac{1}{n_I} \text{Var}\{K_h(X - x_0)\} = O(h^d/n_I).$$

Since $\mu_h \asymp h^d$, Chebyshev's inequality gives (26). $\square$

Lemma 4 (Ratio linearization). *Under Conditions 1 and 2, let a be fixed conditional on the data outside an evaluation fold I , with $n_I/M \rightarrow \rho_I \in (0, 1)$. If the scaled numerator process is tight in $\ell^\infty(\mathcal{Y})$, then*

$$\begin{aligned} & \sqrt{M p_M h^d} \{\widehat{F}_{h,I}^a(y) - F_{h,0}(y)\} \\ &= \sqrt{M p_M h^d} (\mathbb{P}_I - P) \left[\frac{K_h}{\mu_h} \{U_y(a) - F_{h,0}(y)\} \right] + o_{\mathbb{P}}(1) \end{aligned} \quad (27)$$

uniformly over $y \in \mathcal{Y}$.

Proof. The numerator has mean zero, so

$$\widehat{F}_{h,I}^a(y) - F_{h,0}(y) = \frac{\mathbb{P}_I[K_h\{U_y(a) - F_{h,0}(y)\}]}{\mathbb{P}_I K_h}.$$

Write $D_I = \mathbb{P}_I K_h / \mu_h - 1$. By Lemma 3,

$$D_I = O_{\mathbb{P}}\{(n_I h^d)^{-1/2}\} = o_{\mathbb{P}}(1).$$

The assumed tightness and $n_I/M \rightarrow \rho_I$ give

$$\left\| (\mathbb{P}_I - P) \left[\frac{K_h}{\mu_h} \{U(a) - F_{h,0}(\cdot)\} \right] \right\|_{\mathcal{Y}} = O_{\mathbb{P}}\{(M p_M h^d)^{-1/2}\}.$$

Expanding $(1 + D_I)^{-1} = 1 - D_I + o_{\mathbb{P}}(|D_I|)$, the scaled remainder is $O_{\mathbb{P}}(|D_I|) = o_{\mathbb{P}}(1)$, proving (27). $\square$

Conditional on a fitting fold, suppress its index and treat q , m , and D as fixed. Define

$$\mathcal{G}_{M,h} = p_M G_{M,h}, \quad \mathcal{H}_{M,h} = p_M H_{M,h}.$$

Lemma 5 (Observable alignment identity). *Conditional on the candidate regressions,*

$$G_{M,h} = E \left[v_{M,h} \left\langle D, \frac{R}{\pi_M} (Z - q) \right\rangle_v \right]. \quad (28)$$

Proof. Conditional on W ,

$$E \left\{ \frac{R}{\pi_M(W)} (Z_y - q_y(X)) \mid W \right\} = m_{0,y}(W) - q_y(X) = A_y(W).$$

Integrating against $D_y d\nu(y)$, multiplying by $v_{M,h}$, and taking expectation proves (28). $\square$

Lemma 6 (Gain-estimation rates). *Under Conditions 1–3, on a selection fold of size n_G with $n_G/M \rightarrow \rho_G \in (0, 1)$,*

$$\widehat{\mathcal{G}}_h - \mathcal{G}_{M,h} = O_{\mathbb{P}}\{(M p_M h^d)^{-1/2}\}, \quad (29)$$

$$\widehat{\mathcal{H}}_h - \mathcal{H}_{M,h} = O_{\mathbb{P}}\{(M h^d)^{-1/2}\}. \quad (30)$$

The same rates hold when μ_h in $v_{M,h}$ is replaced by the selection-fold kernel mean.

Proof. Let

$$T_{G,M} = p_M v_{M,h} \left\langle D, \frac{R}{\pi_M} (Z - q) \right\rangle_\nu.$$

Uniform boundedness of the paths and finiteness of ν imply, on the support of K_h ,

$$|T_{G,M}| \leq \frac{CR}{p_M h^d}.$$

Since

$$\Pr\{R = 1, K_h(X - x_0) \neq 0\} = E[\pi_M(W) 1\{K_h(X - x_0) \neq 0\}] = O(p_M h^d),$$

we obtain

$$E(T_{G,M}^2) = O\{(p_M h^d)^{-1}\}.$$

An empirical mean over $n_G \asymp M$ observations therefore has standard deviation $O\{(M p_M h^d)^{-1/2}\}$.

For

$$T_{H,M} = p_M v_{M,h} \|D\|_\nu^2,$$

we have $|T_{H,M}| \leq C h^{-d}$ on the kernel support, so

$$E(T_{H,M}^2) = O(h^{-d}).$$

This gives (30). Lemma 3 shows that replacing μ_h^{-2} by its selection-fold counterpart multiplies each criterion term by

$$1 + O_{\mathbb{P}}\{(M h^d)^{-1/2}\}.$$

The normalized population criteria are $O(1)$, and $(M h^d)^{-1/2} \leq (M p_M h^d)^{-1/2}$, so this substitution is absorbed by the displayed rates. $\square$

Lemma 7 (Quadratic regret with numerical stabilization). *Under the conditions of Lemma 6 and*

the stabilizer clauses of Condition 3, let

$$Q_{M,h}(\lambda) = -2\lambda\mathcal{G}_{M,h} + \lambda^2\mathcal{H}_{M,h}$$

and define the stabilized empirical criterion

$$\widehat{Q}_h(\lambda) = -2\lambda\widehat{\mathcal{G}}_h + \lambda^2 \max\{\widehat{\mathcal{H}}_h, \varepsilon_M\}.$$

Let $\widehat{\lambda}_h$ and λ_h^{or} minimize these criteria over $[0, 1]$, and put

$$r_{M,h} = (Mp_M h^d)^{-1/2}.$$

Then

$$\sup_{\lambda \in [0,1]} |\widehat{Q}_h(\lambda) - Q_{M,h}(\lambda)| = O_{\mathbb{P}}(r_{M,h} + \varepsilon_M), \quad (31)$$

$$Q_{M,h}(\widehat{\lambda}_h) - Q_{M,h}(\lambda_h^{\text{or}}) = O_{\mathbb{P}}(r_{M,h} + \varepsilon_M). \quad (32)$$

If $\mathcal{H}_{M,h} \rightarrow H_{\infty} > 0$, then

$$\widehat{\lambda}_h - \lambda_h^{\text{or}} = O_{\mathbb{P}}(r_{M,h}). \quad (33)$$

If, in addition, $\lambda_h^{\text{or}} \in [\delta, 1 - \delta]$ for some $\delta > 0$ and all sufficiently large M , then

$$Q_{M,h}(\widehat{\lambda}_h) - Q_{M,h}(\lambda_h^{\text{or}}) = O_{\mathbb{P}}(r_{M,h}^2). \quad (34)$$

If $\mathcal{H}_{M,h} \rightarrow 0$ and $\sup_M \mathcal{J}_{M,h}(0) < \infty$, then

$$\sup_{\lambda \in [0,1]} |Q_{M,h}(\lambda) - Q_{M,h}(0)| = o(1). \quad (35)$$

Proof. For $0 \leq \lambda \leq 1$,

$$\begin{aligned} & |\widehat{Q}_h(\lambda) - Q_{M,h}(\lambda)| \\ & \leq 2|\widehat{\mathcal{G}}_h - \mathcal{G}_{M,h}| + |\widehat{\mathcal{H}}_h - \mathcal{H}_{M,h}| + \varepsilon_M. \end{aligned}$$

Lemma 6 proves (31). Empirical optimality gives

$$Q_{M,h}(\widehat{\lambda}_h) - Q_{M,h}(\lambda_h^{\text{or}}) \leq 2 \sup_{\lambda \in [0,1]} |\widehat{Q}_h(\lambda) - Q_{M,h}(\lambda)|,$$

which proves (32).

Suppose $\mathcal{H}_{M,h} \rightarrow H_\infty > 0$. Then $\widehat{\mathcal{H}}_h \rightarrow_{\mathbb{P}} H_\infty$, so the stabilizer is inactive with probability tending to one. On that event,

$$\widehat{\lambda}_h = \Pi_{[0,1]} \left(\frac{\widehat{\mathcal{G}}_h}{\widehat{\mathcal{H}}_h} \right).$$

Projection is nonexpansive and the ratio map is Lipschitz in a neighborhood of H_∞ , which gives (33).

If $\lambda_h^{\text{or}} \in [\delta, 1 - \delta]$, then the population minimizer is interior and $\mathcal{G}_{M,h} = \mathcal{H}_{M,h} \lambda_h^{\text{or}}$. Equation (33) places $\widehat{\lambda}_h$ in $(0, 1)$ with probability tending to one. The exact quadratic identity yields

$$Q_{M,h}(\widehat{\lambda}_h) - Q_{M,h}(\lambda_h^{\text{or}}) = \mathcal{H}_{M,h}(\widehat{\lambda}_h - \lambda_h^{\text{or}})^2,$$

which is $O_{\mathbb{P}}(r_{M,h}^2)$.

Now suppose $\mathcal{H}_{M,h} \rightarrow 0$. Weighted Cauchy–Schwarz gives

$$|\mathcal{G}_{M,h}|^2 \leq \mathcal{J}_{M,h}(0) \mathcal{H}_{M,h},$$

so $\mathcal{G}_{M,h} \rightarrow 0$. Uniformly over $\lambda \in [0, 1]$,

$$|Q_{M,h}(\lambda) - Q_{M,h}(0)| \leq 2|\mathcal{G}_{M,h}| + \mathcal{H}_{M,h} = o(1),$$

which proves (35). □

Lemma 8 (Primitive norm comparison). *Under Conditions 1 and 2, suppose ν has a density bounded below on $\mathcal{Y}$, and every path $D_M(W)$ is uniformly α -Hölder with constant L_D :*

$$|D_{M,z}(W) - D_{M,y}(W)| \leq L_D |z - y|^\alpha.$$

Then Condition 6 holds with

$$\omega(t) = Ct^{2\alpha/(2\alpha+1)}.$$

Proof. Let f be an α -Hölder function with constant L_D , and set $a = \|f\|_\infty$. On an interval of length proportional to $a^{1/\alpha}$ around a maximizer of $|f|$, one has $|f| \geq a/2$. Since the density of ν is bounded below,

$$\|f\|_\nu^2 \geq Ca^{2+1/\alpha}, \quad \|f\|_\infty^2 \leq C\{\|f\|_\nu^2\}^\gamma, \quad \gamma = \frac{2\alpha}{2\alpha+1}.$$

Let $w_{M,h}(W) = p_M \nu_{M,h}(W)$. Then

$$\mathcal{D}_{M,h}^2 \leq E\{w_{M,h}\|D_M\|_\infty^2\} \leq CE\{w_{M,h}(\|D_M\|_\nu^2)^\gamma\}.$$

Because $\gamma \in (0, 1)$, Jensen's inequality under the probability measure with density $w_{M,h}/Ew_{M,h}$ gives

$$E\{w_{M,h}(\|D_M\|_\nu^2)^\gamma\} \leq (Ew_{M,h})^{1-\gamma} \{E(w_{M,h}\|D_M\|_\nu^2)\}^\gamma.$$

Conditions 1–2 imply $Ew_{M,h} = O(1)$, while the last expectation is $\mathcal{H}_{M,h}$. Hence

$$\mathcal{D}_{M,h}^2 \leq C\mathcal{H}_{M,h}^{2\alpha/(2\alpha+1)} + o(1),$$

which proves the claim. □

6 Path compatibility on a finite threshold grid

The continuum oracle-equivalence theorem uses Condition 6. On a fixed threshold grid, the corresponding bound follows directly from the discrete transfer criterion.

Let

$$v_L = \sum_{j=1}^L w_j \delta_{y_j}, \quad w_j > 0, \quad \sum_{j=1}^L w_j < \infty,$$

and write $w_{\min} = \min_{1 \leq j \leq L} w_j$. Define

$$\begin{aligned} \mathcal{H}_{M,h,L} &= p_M E \left[v_{M,h}(W) \sum_{j=1}^L w_j D_{M,y_j}^2(W) \right], \\ \mathcal{D}_{M,h,L}^2 &= \max_{1 \leq j \leq L} \frac{p_M h^d}{\mu_h^2} E \left[K_h^2 \left(\frac{1}{\pi_M} - 1 \right) D_{M,y_j}^2 \right]. \end{aligned}$$

Lemma 9 (Finite-grid path compatibility). *For every fixed positive-weight grid,*

$$\mathcal{D}_{M,h,L}^2 \leq \frac{\mathcal{H}_{M,h,L}}{w_{\min}}. \quad (36)$$

The same inequality holds conditionally for each fitted correction path. Consequently, if $\mathcal{H}_{M,h,L} \rightarrow 0$, coefficient perturbations are negligible uniformly over $\{y_1, \dots, y_L\}$, without a continuum Hölder condition.

Proof. For every j , nonnegativity of the local weight gives

$$\mathcal{H}_{M,h,L} \geq w_j \frac{p_M h^d}{\mu_h^2} E \left[K_h^2 \left(\frac{1}{\pi_M} - 1 \right) D_{M,y_j}^2 \right].$$

Divide by w_j , take the maximum over j , and use $w_j \geq w_{\min}$. The fitted-path inequality is identical conditional on the fitting sample. The coefficient-perturbation argument of Lemma 12 then applies with the grid maximum in place of the continuum supremum. $\square$

For the uniform grid used in computation, $w_j = 1/L$ and (36) becomes

$$\mathcal{D}_{M,h,L}^2 \leq L\mathcal{H}_{M,h,L}.$$

The growing-grid result in Proposition 10 connects these finite-grid processes with the continuum process when the grid mesh tends to zero.

7 Normalization of the transfer weight

The population criterion uses

$$v_{M,h}(W) = \frac{h^d K_h^2(X - x_0)}{\mu_h^2} \left\{ \frac{1}{\pi_M(W)} - 1 \right\}.$$

The coefficient and transfer sign are invariant to a positive normalization that is common to every observation and threshold.

Lemma 10 (Weight-normalization invariance). *Let $c_{M,h} > 0$ and replace $v_{M,h}$ by $v_{M,h}^c = c_{M,h}v_{M,h}$.*

Then

$$J_{M,h}^c(\lambda) = c_{M,h}J_{M,h}(\lambda), \quad G_{M,h}^c = c_{M,h}G_{M,h}, \quad H_{M,h}^c = c_{M,h}H_{M,h}.$$

Consequently,

$$\Pi_{[0,1]}(G_{M,h}^c/H_{M,h}^c) = \lambda_h^{\text{or}},$$

and

$$\text{sign}(2G_{M,h}^c - H_{M,h}^c) = \text{sign}(2G_{M,h} - H_{M,h}).$$

The same identities hold for the selection-fold empirical criterion under a positive fold-specific constant.

Proof. Each criterion quantity is linear in the weight. The ratio and sign conclusions follow because $c_{M,h} > 0$. □

Taking

$$c_{M,h} = \mu_h^2 / h^{2d}$$

produces the equivalent population weight

$$h^{-d} K_h^2(X - x_0) \left\{ \frac{1}{\pi_M(W)} - 1 \right\}.$$

On a selection fold, the implementation uses

$$\widehat{v}_h(W) = \frac{h^d K_h^2(X - x_0)}{\widehat{\mu}_h^2} \left\{ \frac{1}{\pi_M(W)} - 1 \right\}.$$

This equals the empirical version of the equivalent h^{-d} weight above multiplied by the positive fold-specific constant $h^{2d} / \widehat{\mu}_h^2$. The constant changes the numerical scale of $\widehat{G}_h$ and $\widehat{H}_h$, but not their ratio or the transfer sign.

8 Path-oracle equivalence under sample splitting

Let $\mathcal{T}_k$ collect the fitting-fold and selection-fold data associated with evaluation fold k . Write $\widehat{F}_{h,k}(b)$ for the evaluation-fold estimator formed with augmentation path b .

Lemma 11 (Regression-path perturbation). *Under Conditions 1–5,*

$$\sqrt{N_{M,h}} \left\| \widehat{F}_{h,k}(a_k^{\text{or}}) - \widehat{F}_{h,k}(a_M) \right\|_{\mathcal{Y}} = o_{\mathbb{P}}(1),$$

provided

$$\|a_k^{\text{or}} - a_M\|_{\text{ver}} = o_{\mathbb{P}}(1)$$

and the realized difference class satisfies Condition 5.

Proof. For every y ,

$$U_y(a_k^{\text{or}}) - U_y(a_M) = \{a_{k,y}^{\text{or}} - a_{M,y}\} \left(1 - \frac{R}{\pi_M} \right). \quad (37)$$

Conditional on $\mathcal{T}_k$, the right side has conditional mean zero given W . Define

$$\mathbb{D}_{M,k,y} = \sqrt{p_M h^d} \frac{K_h}{\mu_h} \{a_{k,y}^{\text{or}} - a_{M,y}\} \left(1 - \frac{R}{\pi_M}\right).$$

Its conditional squared $L_2(P)$ norm is

$$\frac{p_M h^d}{\mu_h^2} E \left[K_h^2 \left(\frac{1}{\pi_M} - 1 \right) \{a_{k,y}^{\text{or}} - a_{M,y}\}^2 \mid \mathcal{T}_k \right],$$

which is uniformly $o_{\mathbb{P}}(1)$. The conditional maximal inequality for VC-type classes gives

$$\left\| \frac{1}{\sqrt{|I_{E,k}|}} \sum_{i \in I_{E,k}} \mathbb{D}_{M,k,\cdot}(O_i) \right\|_{\mathcal{Y}} = o_{\mathbb{P}}(1).$$

Because $|I_{E,k}|/M \rightarrow \rho_k \in (0, 1)$, replacing $\sqrt{|I_{E,k}|}$ by $\sqrt{M}$ changes only a bounded deterministic factor. Lemma 4 then gives the stated estimator difference. $\square$

Lemma 12 (Coefficient perturbation). *Under Conditions 1–6,*

$$\sqrt{N_{M,h}} \left\| \widehat{F}_{h,k}(\widehat{a}_k) - \widehat{F}_{h,k}(a_k^{\text{or}}) \right\|_{\mathcal{Y}} = o_{\mathbb{P}}(1),$$

where

$$a_{k,y}^{\text{or}} = \widehat{q}_{k,y} + \lambda_{h,k}^{\text{or}} \widehat{D}_{k,y}, \quad \widehat{D}_{k,y} = \widehat{m}_{k,y} - \widehat{q}_{k,y},$$

uses the same fitted regressions as $\widehat{a}_k$ and replaces only the estimated coefficient by its conditional population minimizer.

Proof. Conditional on $\mathcal{T}_k$,

$$U_y(\widehat{a}_k) - U_y(a_k^{\text{or}}) = (\widehat{\lambda}_{h,k} - \lambda_{h,k}^{\text{or}}) \widehat{D}_{k,y}(W) \left(1 - \frac{R}{\pi_M(W)}\right). \quad (38)$$

The indexed factor following the coefficient difference is a centered local empirical process. Let

$\widehat{\mathcal{D}}_{h,k}^2$ be its fitted squared maximal local L_2 radius. Condition 6 gives

$$\widehat{\mathcal{D}}_{h,k}^2 \leq \omega(\mathcal{H}_{h,k}) + o_{\mathbb{P}}(1).$$

The conditional maximal inequality and Lemma 4 yield

$$\sqrt{N_{M,h}} \left\| \widehat{F}_{h,k}(\widehat{a}_k) - \widehat{F}_{h,k}(a_k^{\text{or}}) \right\|_{\mathcal{Y}} = O_{\mathbb{P}} \left(|\widehat{\lambda}_{h,k} - \lambda_{h,k}^{\text{or}}| \widehat{\mathcal{D}}_{h,k} \right) + o_{\mathbb{P}}(1).$$

If $\mathcal{H}_{h,k} \rightarrow H_{\infty,k} > 0$, Lemma 7 gives

$$|\widehat{\lambda}_{h,k} - \lambda_{h,k}^{\text{or}}| = o_{\mathbb{P}}(1),$$

and $\widehat{\mathcal{D}}_{h,k} = O_{\mathbb{P}}(1)$. If $\mathcal{H}_{h,k} \rightarrow 0$, Condition 6 gives $\widehat{\mathcal{D}}_{h,k} \rightarrow_{\mathbb{P}} 0$, while the coefficient difference is bounded by one. The product is $o_{\mathbb{P}}(1)$ in both regimes. $\square$

Proof of Theorem 1. The gain-estimation rates and criterion results follow from Lemmas 6 and 7. Lemma 12 replaces the estimated coefficient by the fold-specific conditional population minimizer while holding the fitted regressions fixed. Weighted summation over the three evaluation folds preserves the $o_{\mathbb{P}}(1)$ order. Lemma 11 then replaces the fitted population-minimizing paths by the common triangular path a_M . The positive-curvature coefficient rate and the interior quadratic regret are the corresponding statements of Lemma 7; the vanishing-curvature result follows from Condition 6. $\square$

9 Uniform local conditional-distribution process

Throughout the Supplement, $\Rightarrow$ denotes weak convergence and $\Rightarrow_{\mathbb{P}}$ denotes conditional weak convergence in probability.

Define the path-oracle triangular influence process

$$\Psi_{M,y}(O) = \sqrt{p_M h^d} \frac{K_h(X - x_0)}{\mu_h} \{U_y(a_M) - F_{h,0}(y \mid x_0)\}. \quad (39)$$

By Lemma 1, $P\Psi_{M,y} = 0$.

Lemma 13 (Asymptotic linear representation). *Under Conditions 1–6,*

$$\mathbb{Z}_M(y) = \frac{1}{\sqrt{M}} \sum_{i=1}^M \Psi_{M,y}(O_i) + o_{\mathbb{P}}(1) \quad (40)$$

uniformly over $y \in \mathcal{Y}$, where $\mathbb{Z}_M$ is defined in (6).

Proof. Theorem 1 replaces the estimated coefficient by the conditional population minimizer, and Lemma 11 replaces each fitted fold-specific augmentation by a_M , with a uniform $o_{\mathbb{P}}(1)$ error after scaling. Lemma 4 linearizes each fold-specific ratio. If $n_k = |I_{E,k}|$ and $\rho_{M,k} = n_k/M$, then

$$\rho_{M,k} \frac{1}{n_k} \sum_{i \in I_{E,k}} f(O_i) = \frac{1}{M} \sum_{i \in I_{E,k}} f(O_i).$$

The disjoint evaluation folds therefore combine into the full-sample empirical average in (40). $\square$

Lemma 14 (Covariance limit). *Under Conditions 1, 2, and 7,*

$$E\{\Psi_{M,y}\Psi_{M,z}\} \rightarrow \Gamma_p(y, z)$$

uniformly on $\mathcal{Y}^2$, where

$$\begin{aligned} \Gamma_p(y, z) = & \frac{\kappa_K}{f_X(x_0)} E \left[\frac{C_{yz}(W)}{e_0(W)} + \left\{ \frac{1}{e_0(W)} - p \right\} \Delta_y(W) \Delta_z(W) \right. \\ & \left. + p \{m_{0,y}(W) - F_0(y \mid x_0)\} \{m_{0,z}(W) - F_0(z \mid x_0)\} \mid X = x_0 \right]. \quad (41) \end{aligned}$$

Proof. Conditional on W , Lemmas 2 and 1 give

$$\begin{aligned} E \left[\{U_y(a_M) - F_{h,0}(y)\} \{U_z(a_M) - F_{h,0}(z)\} \mid W \right] \\ = \frac{C_{yz}(W)}{\pi_M(W)} + \left\{ \frac{1}{\pi_M(W)} - 1 \right\} \Delta_{M,y}(W) \Delta_{M,z}(W) \\ + \{m_{0,y}(W) - F_{h,0}(y)\} \{m_{0,z}(W) - F_{h,0}(z)\}. \end{aligned}$$

Multiplying by p_M and using $\pi_M = p_M e_M$ gives $B_{M,yz}(W)$ in Condition 7. Consequently,

$$E \{ \Psi_{M,y} \Psi_{M,z} \} = \frac{h^d}{\mu_h^2} E \{ K_h^2(X - x_0) B_{M,yz}(W) \}.$$

Changing variables $x = x_0 + hu$ writes this expression as

$$\frac{\int K^2(u) f_X(x_0 + hu) E \{ B_{M,yz}(W) \mid X = x_0 + hu \} du}{\left[\int K(u) f_X(x_0 + hu) du \right]^2}.$$

Condition 7, continuity and positivity of f_X , and dominated convergence yield (41) uniformly on $\mathcal{Y}^2$. □

Lemma 15 (Increment semimetric). *Under Conditions 1, 2, and 5,*

$$\| \Psi_{M,z} - \Psi_{M,y} \|_{P,2}^2 \leq C \{ |z - y| + |z - y|^{2\alpha} \} \quad (42)$$

uniformly in M and $y, z \in \mathcal{Y}$.

Proof. For $y < z$,

$$U_z(a_M) - U_y(a_M) = \{a_{M,z} - a_{M,y}\} \left(1 - \frac{R}{\pi_M} \right) + \frac{R}{\pi_M} 1(y < Y \leq z). \quad (43)$$

Using $(r + s)^2 \leq 2r^2 + 2s^2$, conditional expectation, and $R^2 = R$,

$$p_M E [\{U_z - U_y\}^2 \mid W] \leq C |a_{M,z} - a_{M,y}|^2 + C \Pr(y < Y \leq z \mid W).$$

Conditions (5)–(6) provide the threshold modulus. In addition,

$$|F_{h,0}(z \mid x_0) - F_{h,0}(y \mid x_0)| \leq C_{\mathcal{Y}}|z - y|$$

by the same conditional-CDF increment bound and the nonnegative kernel weights. Since

$$\frac{h^d E K_h^2}{\mu_h^2} = O(1),$$

the centered increment satisfies (42). □

Lemma 16 (Triangular empirical-process central limit theorem). *Under Conditions 1, 2, 5, and 7,*

$$\frac{1}{\sqrt{M}} \sum_{i=1}^M \Psi_{M,\cdot}(O_i) \Rightarrow \mathbb{G}_p \quad \text{in } \ell^\infty(\mathcal{Y}),$$

where $\mathbb{G}_p$ has covariance kernel Γ_p .

Proof. Let $\mathcal{F}_M = \{\Psi_{M,y} : y \in \mathcal{Y}\}$ and let F_M be its envelope. Boundedness of a_M , compact support and boundedness of K , and $\pi_M \geq p_M c_e$ give

$$\|F_M\|_\infty \leq \frac{C}{\sqrt{p_M h^d}}, \quad \sup_M P F_M^2 < \infty. \quad (44)$$

Because $M p_M h^d \rightarrow \infty$,

$$P\{F_M^2 1(F_M > \eta \sqrt{M})\} = 0 \quad \text{for every } \eta > 0 \text{ and all sufficiently large } M. \quad (45)$$

Thus the row-wise Lindeberg condition holds.

For fixed thresholds $y_1, \dots, y_L$ and constants $c_1, \dots, c_L$, set

$$S_M = \sum_{j=1}^L c_j \Psi_{M,y_j}.$$

Equation (45) gives Lindeberg's condition for S_M , and Lemma 14 gives convergence of its variance.

The Lindeberg–Feller theorem and Cramér–Wold yield finite-dimensional convergence.

For equicontinuity, define

$$\rho(y, z) = C\{|y - z| + |y - z|^{2\alpha}\}^{1/2}$$

with C chosen so that $\|\Psi_{M,z} - \Psi_{M,y}\|_{P,2} \leq \rho(y, z)$. The compact interval $\mathcal{Y}$ is totally bounded under ρ . For $\delta > 0$, let

$$\mathcal{F}_M(\delta) = \{\Psi_{M,z} - \Psi_{M,y} : \rho(y, z) \leq \delta\}.$$

This difference class is VC-type with

$$\sup_{f \in \mathcal{F}_M(\delta)} \|f\|_{P,2} \leq \delta.$$

The VC entropy integral satisfies

$$J(\delta, \mathcal{F}_M) = \sup_Q \int_0^\delta \sqrt{1 + \log N\{\epsilon \|F_M\|_{Q,2}, \mathcal{F}_M, L_2(Q)\}} d\epsilon \leq C\delta \sqrt{\log(A/\delta)}.$$

The maximal inequality for measurable VC-type classes in Section 2.14 of van der Vaart and Wellner (1996) therefore gives

$$E \left\| \frac{1}{\sqrt{M}} \sum_{i=1}^M \{f(O_i) - Pf\} \right\|_{\mathcal{F}_M(\delta)} \leq C\delta \sqrt{\log(A/\delta)} + C \frac{\|F_M\|_\infty}{\sqrt{M}} \log(A/\delta). \quad (46)$$

For fixed δ , the second term tends to zero by (44); the first tends to zero as $\delta \downarrow 0$. Markov's inequality gives asymptotic ρ -equicontinuity. Finite-dimensional convergence, total boundedness, the Lindeberg condition, and this entropy bound verify the triangular-array functional central limit criteria developed by Ziegler (1997) and Arcones (1999). Hence the process converges weakly in $\ell^\infty(\mathcal{Y})$. $\square$

Proof of Theorem 2. Lemma 13 gives the uniform asymptotic linear representation. Lemma 16 gives weak convergence of the leading term, and Lemma 14 identifies the covariance kernel.

Slutsky's theorem in $\ell^\infty(\mathcal{Y})$ completes the proof. $\square$

Corollary 1 (Finite-threshold inference). *Fix $L < \infty$, thresholds $y_1 < \dots < y_L$, and a positive transfer measure $\nu_L = \sum_{j=1}^L w_j \delta_{y_j}$, where $w_j > 0$. Define*

$$\begin{aligned}\mathbf{F}_{h,0} &= \{F_{h,0}(y_1 | x_0), \dots, F_{h,0}(y_L | x_0)\}^\top, \\ \widehat{\mathbf{F}}_h &= \{\widehat{F}_h^{\text{raw}}(y_1 | x_0), \dots, \widehat{F}_h^{\text{raw}}(y_L | x_0)\}^\top.\end{aligned}$$

Under Conditions 1, 2, 3, 4, and 6, together with finite-dimensional covariance convergence,

$$\sqrt{N_{M,h}}\{\widehat{\mathbf{F}}_h - \mathbf{F}_{h,0}\} \Rightarrow N_L(0, \mathbf{\Gamma}_p),$$

where the (j, ℓ) entry of $\mathbf{\Gamma}_p$ is $\Gamma_p(y_j, y_\ell)$. The corresponding Rademacher multiplier vector consistently estimates this law. The result applies to continuous, discrete, and mixed outcomes.

Proof. Fix $\mathcal{Y}_L = \{y_1, \dots, y_L\}$. Lemma 6 gives the gain-estimation rates on the fixed grid. In the positive-curvature regime, Lemma 7 gives $\widehat{\lambda}_{h,k} - \lambda_{h,k}^{\text{or}} = o_{\mathbb{P}}(1)$; in the vanishing-curvature regime, Lemma 9 makes the correction radius vanish on $\mathcal{Y}_L$. The same two-regime argument as in Lemma 12 therefore replaces the estimated coefficient by its conditional population minimizer on the grid. Condition 4 and coordinatewise Chebyshev bounds replace the fitted population-minimizing paths by a_M , without a continuum entropy condition. Lemma 4 then gives a finite-dimensional asymptotic linear representation.

The envelope bound (44) implies Lindeberg's condition, and covariance convergence at (y_j, y_ℓ) gives the limiting covariance matrix. Cramér–Wold proves the Gaussian vector limit. For the multiplier vector, coordinatewise influence-function replacement, empirical covariance convergence, and the conditional multivariate Lindeberg–Feller theorem give the same limit. No threshold-density or continuum equicontinuity condition enters this result. $\square$

10 Limit of the local efficiency bound

Corollary 2 (Limit of the local efficiency bound). *Suppose the conditions of Theorem 2 hold and the oracle version of Condition 7 holds for the canonical-gradient covariance. Define*

$$\begin{aligned} \Gamma_p^{\text{eff}}(y, z) = & \frac{\kappa_K}{f_X(x_0)} E \left[\frac{C_{yz}(W)}{e_0(W)} \right. \\ & \left. + p \{m_{0,y}(W) - F_0(y \mid x_0)\} \{m_{0,z}(W) - F_0(z \mid x_0)\} \mid X = x_0 \right]. \end{aligned} \quad (47)$$

For every finite threshold vector $\mathbf{y}$,

$$p_M h^d \Sigma_{M,h}^{\text{eff}}(\mathbf{y}) \longrightarrow \mathbf{\Gamma}_p^{\text{eff}}(\mathbf{y}) \quad (48)$$

entrywise. Moreover,

$$\Gamma_p(y, z) - \Gamma_p^{\text{eff}}(y, z) = \frac{\kappa_K}{f_X(x_0)} E \left[\left\{ \frac{1}{e_0(W)} - p \right\} \Delta_y(W) \Delta_z(W) \mid X = x_0 \right], \quad (49)$$

and this difference is positive semidefinite over every finite threshold collection.

Proof. For thresholds y, z , the conditional second moment of the canonical gradient in (13) is

$$\begin{aligned} & E \left[\{U_y(m_0) - F_{h,0}(y \mid x_0)\} \{U_z(m_0) - F_{h,0}(z \mid x_0)\} \mid W \right] \\ &= \frac{C_{yz}(W)}{\pi_M(W)} + \{m_{0,y}(W) - F_{h,0}(y \mid x_0)\} \{m_{0,z}(W) - F_{h,0}(z \mid x_0)\}. \end{aligned}$$

Therefore

$$\begin{aligned} & p_M h^d E \{ \phi_{M,h,y}^{\text{eff}} \phi_{M,h,z}^{\text{eff}} \} \\ &= \frac{h^d}{\mu_h^2} E \left[K_h^2(X - x_0) \left\{ \frac{C_{yz}(W)}{e_M(W)} \right. \right. \\ & \quad \left. \left. + p_M \{m_{0,y}(W) - F_{h,0}(y \mid x_0)\} \{m_{0,z}(W) - F_{h,0}(z \mid x_0)\} \right\} \right]. \end{aligned} \quad (50)$$

The primitive covariance conditions in Proposition 1, applied with zero augmentation error, and the kernel change of variables used in Lemma 14 show that (50) converges to $\Gamma_p^{\text{eff}}(y, z)$. Applying this entrywise gives (48).

Subtracting (47) from (8) gives (49). Since $\pi_M = p_M e_M \leq 1$, every local limit satisfies $p e_0(W) \leq 1$, and hence

$$\frac{1}{e_0(W)} - p \geq 0.$$

For thresholds $y_1, \dots, y_L$ and $c \in \mathbb{R}^L$, the quadratic form of the difference matrix is

$$\frac{\kappa_K}{f_X(x_0)} E \left[\left\{ \frac{1}{e_0(W)} - p \right\} \left\{ \sum_{j=1}^L c_j \Delta_{y_j}(W) \right\}^2 \mid X = x_0 \right] \geq 0.$$

This proves the positive-semidefinite ordering. □

11 Inference under scarce reference sampling

The normalization

$$p_M = E\{\pi_M(W)\}, \quad e_M(W) = \pi_M(W)/p_M$$

separates the reference budget from its allocation. It preserves the borrowing coefficient because

$$\frac{p_M G_{M,h}}{p_M H_{M,h}} = \frac{G_{M,h}}{H_{M,h}},$$

and it preserves the transfer sign because

$$p_M \{2G_{M,h} - H_{M,h}\} = 2\mathcal{G}_{M,h} - \mathcal{H}_{M,h}.$$

Proposition 2 (Scarce-reference covariance). *Under Theorem 2, if $p_M \rightarrow 0$, then*

$$\mathbb{Z}_M \Rightarrow \mathbb{G}_0 \quad \text{in } \ell^\infty(\mathcal{Y}),$$

with covariance kernel

$$\Gamma_0(y, z) = \frac{\kappa_K}{f_X(x_0)} E \left[\frac{C_{yz}(W) + \Delta_y(W)\Delta_z(W)}{e_0(W)} \mid X = x_0 \right]. \quad (51)$$

Consequently,

$$\text{Var}\{\widehat{F}_h(y \mid x_0)\} = \frac{\Gamma_0(y, y) + o(1)}{Mp_M h^d}.$$

Proof. Set $p = 0$ in (41). The residual-heterogeneity term is multiplied by p and vanishes, leaving (51). The variance expansion follows from Lemma 13. $\square$

The quantity

$$N_{\text{eff}}(x_0, h) = Mp_M h^d$$

is the first-order local reference size. The allocation function $e_0(W)$ changes the covariance constant even when two studies have the same total reference count Mp_M .

11.1 Target-specific allocation of the reference budget

For a fixed augmentation path a , define the integrated local residual variation

$$V_a(W) = \int_{\mathcal{Y}} [C_{yy}(W) + \{\delta_y^a(W)\}^2] d\nu(y), \quad \delta_y^a = a_y - m_{0,y}. \quad (52)$$

The allocation-dependent part of the limiting integrated variance is proportional to

$$E\{V_a(W)/e_0(W) \mid X = x_0\}.$$

Proposition 3 (Locally optimal verification intensity). *Fix x_0 , an augmentation path a , and a local budget*

$$E\{e_0(W) \mid X = x_0\} = \bar{e}(x_0) > 0.$$

Assume

$$0 < V_a(W) < \infty$$

almost surely under the local law $W \mid X = x_0$, and

$$0 < E\{\sqrt{V_a(W)} \mid X = x_0\} < \infty.$$

Among positive measurable intensities satisfying the local budget, the unique minimizer is

$$e_0^{\text{opt}}(W) = \bar{e}(x_0) \frac{\sqrt{V_a(W)}}{E\{\sqrt{V_a(W)} \mid X = x_0\}}, \quad X = x_0. \quad (53)$$

The minimum allocation-dependent variance is

$$\frac{\{E[\sqrt{V_a(W)} \mid X = x_0]\}^2}{\bar{e}(x_0)}. \quad (54)$$

Suppose instead that $c_e \leq e_0(W) \leq C_e$ and $c_e \leq \bar{e}(x_0) \leq C_e$. The constrained problem has a unique minimizer. If $c_e < \bar{e}(x_0) < C_e$, it has the form

$$e_0^{\text{opt}}(W) = \min\{C_e, \max(c_e, c\sqrt{V_a(W)})\}, \quad (55)$$

where $c > 0$ is chosen so that the budget holds. Every such multiplier produces the same minimizing intensity almost surely. At the endpoint budgets $\bar{e}(x_0) = c_e$ and $\bar{e}(x_0) = C_e$, the minimizers are the corresponding constant intensities.

Proof. All expectations in this proof are conditional on $X = x_0$. Cauchy–Schwarz gives

$$E\left\{\frac{V_a(W)}{e_0(W)}\right\} E\{e_0(W)\} \geq \{E[\sqrt{V_a(W)}]\}^2.$$

Equality holds exactly when $e_0(W)$ is proportional to $\sqrt{V_a(W)}$. The budget constraint determines the proportionality constant in (53), and strict convexity of v/e in $e > 0$ for every $v > 0$ gives

uniqueness up to null sets.

Under box constraints, the integral objective is strictly convex on the feasible set and therefore has a unique minimizing intensity. The Karush–Kuhn–Tucker conditions give

$$e_0^{\text{opt}}(W) = \min\{C_e, \max(c_e, c\sqrt{V_a(W)})\}$$

for a positive multiplier c . The map

$$g(c) = E[\min\{C_e, \max(c_e, c\sqrt{V_a(W)})\}]$$

is continuous and nondecreasing, with $g(c) \rightarrow c_e$ as $c \downarrow 0$ and $g(c) \rightarrow C_e$ as $c \uparrow \infty$. Hence a multiplier satisfying the interior budget exists. If more than one multiplier satisfies it, the associated clipped intensities coincide almost surely by uniqueness of the minimizer. The endpoint cases follow directly from feasibility. $\square$

Proposition 3 links surrogate quality and validation design. Verification is concentrated where conditional threshold variation and augmentation error are large. When $a = m_0$, the allocation depends only on irreducible reference-outcome variation; for a learned path, it also targets regions where the surrogate correction remains inaccurate.

11.2 Covariance degeneracy

Proposition 4 (Degenerate process directions). *For a finite signed measure ϑ , let*

$$Z_\vartheta = \int Z_y d\vartheta(y), \quad \Delta_\vartheta = \int \Delta_y d\vartheta(y).$$

In the scarce-reference regime, the limiting variance of the projected process is zero if and only if

$$\text{Var}(Z_\vartheta \mid W) = 0 \quad \text{and} \quad \Delta_\vartheta(W) = 0$$

almost surely under the local law $W \mid X = x_0$. On every subspace with positive projected variance, Theorems 2 and 3 apply without modification.

Proof. Integrating (51) against $d\vartheta(y)d\vartheta(z)$ gives

$$\frac{\kappa_K}{f_X(x_0)} E \left[\frac{\text{Var}(Z_\vartheta \mid W) + \{\Delta_\vartheta(W)\}^2}{e_0(W)} \mid X = x_0 \right].$$

The integrand is nonnegative and e_0 is bounded away from zero. The expectation is zero exactly under the two stated conditions. $\square$

12 Multiplier bootstrap

For observation i in evaluation fold $k(i)$, define

$$\widehat{\Psi}_{i,y} = \sqrt{p_M h^d} \frac{K_h(X_i - x_0)}{\widehat{\mu}_{h,k(i)}} \{U_{i,y}(\widehat{a}_{k(i)}) - \widehat{F}_{h,k(i)}(y)\}.$$

The bootstrap process is

$$\mathbb{G}_M^\xi(y) = \frac{1}{\sqrt{M}} \sum_{i=1}^M \xi_i \widehat{\Psi}_{i,y},$$

where ξ_i are independent Rademacher multipliers, independent of the data.

Lemma 17 (Estimated influence-process replacement). *Under Conditions 1–6,*

$$\sup_{y \in \mathcal{Y}} \frac{1}{M} \sum_{i=1}^M \{\widehat{\Psi}_{i,y} - \Psi_{M,y}(O_i)\}^2 = o_{\mathbb{P}}(1), \quad (56)$$

and the realized difference class obeys the polynomial entropy bound in Condition 5 with an empirical L_2 envelope of order $o_{\mathbb{P}}(1)$.

Proof. Fix an evaluation fold k and suppress the fold index. Put

$$r_{\mu,k} = \frac{\mu_h}{\widehat{\mu}_{h,k}} - 1, \quad r_{F,k,y} = \widehat{F}_{h,k}(y) - F_{h,0}(y).$$

The estimated-minus-path-oracle influence function has the exact decomposition

$$\widehat{\Psi}_{i,y} - \Psi_{M,y}(O_i) = r_{\mu,k} \sqrt{p_M h^d} \frac{K_{h,i}}{\mu_h} \{U_{i,y}(\widehat{a}_k) - \widehat{F}_{h,k}(y)\} \quad (57)$$

$$+ \sqrt{p_M h^d} \frac{K_{h,i}}{\mu_h} \{\widehat{a}_{k,y}(W_i) - a_{M,y}(W_i)\} \left(1 - \frac{R_i}{\pi_M(W_i)}\right) \quad (58)$$

$$- \sqrt{p_M h^d} \frac{K_{h,i}}{\mu_h} r_{F,k,y}, \quad (59)$$

where $K_{h,i} = K_h(X_i - x_0)$.

Lemma 3 gives

$$\max_{k \leq 3} |r_{\mu,k}| = O_{\mathbb{P}}\{(Mh^d)^{-1/2}\} = o_{\mathbb{P}}(1).$$

Conditional on the fitted path, boundedness of $\widehat{a}_k$ and $c_e \leq e_M$ imply

$$\begin{aligned} & E \left[\sup_{y \in \mathcal{Y}} p_M h^d \frac{K_h^2}{\mu_h^2} \{U_y(\widehat{a}_k) - \widehat{F}_{h,k}(y)\}^2 \mid \mathcal{T}_k \right] \\ & \leq C \frac{p_M h^d}{\mu_h^2} E \left[K_h^2 \left\{ 1 + \frac{1}{\pi_M(W)} \right\} \right] = O_{\mathbb{P}}(1). \end{aligned}$$

Conditional Markov inequality gives an $O_{\mathbb{P}}(1)$ empirical second moment for the scaled bracket in (57); multiplication by $r_{\mu,k}^2$ makes this contribution $o_{\mathbb{P}}(1)$.

For (58), conditional expectation given the fitting and selection folds equals

$$\frac{p_M h^d}{\mu_h^2} E \left[K_h^2 \left(\frac{1}{\pi_M} - 1 \right) (\widehat{a}_{k,y} - a_{M,y})^2 \mid \mathcal{T}_k \right].$$

Lemmas 11 and 12, together with Conditions 4 and 6, make its supremum $o_{\mathbb{P}}(1)$. The conditional VC maximal inequality transfers the same order to the empirical second moment uniformly over y .

Lemma 4, tightness of the foldwise numerator processes, and the path-replacement lemmas give

$$\max_{k \leq 3} \sup_{y \in \mathcal{Y}} |r_{F,k,y}| = O_{\mathbb{P}}(N_{M,h}^{-1/2}).$$

Moreover,

$$\frac{p_M h^d}{\mu_h^2} \mathbb{P}_M K_h^2 = O_{\mathbb{P}}(p_M),$$

so the empirical L_2 norm of (59) is $o_{\mathbb{P}}(1)$. Summing the three foldwise bounds proves (56).

The difference class is a finite sum of a scalar multiple of a localized influence class, a localized fitted-path difference class multiplied by $1 - R/\pi_M$, and a threshold-indexed centering class. The closure properties of VC-type classes preserve the polynomial covering bound, and the preceding calculations give an empirical L_2 envelope of order $o_{\mathbb{P}}(1)$. $\square$

Lemma 18 (Empirical covariance consistency). *Under Conditions 1–7,*

$$\sup_{y,z \in \mathcal{Y}} \left| \frac{1}{M} \sum_{i=1}^M \widehat{\Psi}_{i,y} \widehat{\Psi}_{i,z} - \Gamma_p(y, z) \right| = o_{\mathbb{P}}(1). \quad (60)$$

Proof. Let

$$\mathcal{P}_M = \{\Psi_{M,y} \Psi_{M,z} : (y, z) \in \mathcal{Y}^2\}.$$

The product class is VC-type and has envelope F_M^2 . The localization calculation gives

$$\sup_{y \in \mathcal{Y}} P \Psi_{M,y}^4 = O\{(p_M h^d)^{-1}\}, \quad \|F_M^2\|_{P,2} = O\{(p_M h^d)^{-1/2}\}. \quad (61)$$

The maximal inequality in Section 2.14 of van der Vaart and Wellner (1996), applied to $\mathcal{P}_M$, yields

$$\sup_{y,z \in \mathcal{Y}} |(\mathbb{P}_M - P)(\Psi_{M,y} \Psi_{M,z})| = O_{\mathbb{P}}\{(M p_M h^d)^{-1/2} + (M p_M h^d)^{-1}\} = o_{\mathbb{P}}(1). \quad (62)$$

The same VC-type moment regime underlies the Gaussian and bootstrap coupling results of Chernozhukov et al. (2014) and Chernozhukov et al. (2016). Lemma 14 gives

$$\sup_{y,z} |P(\Psi_{M,y} \Psi_{M,z}) - \Gamma_p(y, z)| \rightarrow 0.$$

Set

$$\mathcal{E}_{i,y} = \widehat{\Psi}_{i,y} - \Psi_{M,y}(O_i).$$

Lemma 17 gives

$$\sup_y \mathbb{P}_M \mathcal{E}_y^2 = o_{\mathbb{P}}(1),$$

while (62) and covariance convergence imply

$$\sup_y \mathbb{P}_M \Psi_{M,y}^2 = O_{\mathbb{P}}(1).$$

Uniform Cauchy–Schwarz therefore gives

$$\begin{aligned} & \sup_{y,z} \left| \mathbb{P}_M(\widehat{\Psi}_y \widehat{\Psi}_z - \Psi_y \Psi_z) \right| \\ & \leq 2 \left\{ \sup_y \mathbb{P}_M \mathcal{E}_y^2 \right\}^{1/2} \left(\sup_y \mathbb{P}_M \Psi_{M,y}^2 \right)^{1/2} + \sup_y \mathbb{P}_M \mathcal{E}_y^2 = o_{\mathbb{P}}(1). \end{aligned}$$

Combining the bounds proves (60). □

Lemma 19 (Conditional multiplier convergence). *Under Conditions 1–7,*

$$\frac{1}{\sqrt{M}} \sum_{i=1}^M \xi_i \Psi_{M,\cdot}(O_i) \Rightarrow_{\mathbb{P}} \mathbb{G}_p \quad \text{in } \ell^\infty(\mathcal{Y}).$$

Proof. Write

$$\mathbb{G}_M^{\xi, \text{or}}(y) = M^{-1/2} \sum_{i=1}^M \xi_i \Psi_{M,y}(O_i).$$

Conditional on the observations, the summands are independent and centered. For every finite threshold vector, Lemma 18 gives convergence of the conditional covariance matrix. In addition,

$$\max_{1 \leq i \leq M} \frac{\|\Psi_M(O_i)\|_{\mathcal{Y}}}{\sqrt{M}} \leq \frac{C}{\sqrt{M p_M h^d}} \rightarrow 0,$$

so the conditional Lindeberg condition holds and the conditional finite-dimensional laws converge.

For tightness, define

$$\rho_M^2(y, z) = \mathbb{P}_M \{\Psi_{M,y} - \Psi_{M,z}\}^2.$$

Applying (62) to the squared-increment class and using Lemma 15 gives

$$\rho_M^2(y, z) \leq C\rho^2(y, z) + o_{\mathbb{P}}(1) \quad (63)$$

uniformly in y, z . The conditional covering numbers under $L_2(\mathbb{P}_M)$ obey the polynomial bound in Condition 5. The conditional Rademacher maximal inequality gives

$$E_{\xi} \|\mathbb{G}_M^{\xi, \text{or}}\|_{\rho, \delta} \leq C\delta \sqrt{\log(A/\delta)} + C \frac{\|F_M\|_{\infty}}{\sqrt{M}} \log(A/\delta) + o_{\mathbb{P}}(1). \quad (64)$$

The envelope term vanishes because $M p_M h^d \rightarrow \infty$, and the first term vanishes as $\delta \downarrow 0$. Thus $\mathbb{G}_M^{\xi, \text{or}}$ is conditionally asymptotically equicontinuous. Conditional finite-dimensional convergence, covariance consistency, and equicontinuity verify the triangular-array multiplier conditions of Kosorok (2003); the increasing-complexity multiplier coupling theory of Chernozhukov et al. (2016) gives the same conclusion under the displayed VC-type bounds. Hence $\mathbb{G}_M^{\xi, \text{or}} \Rightarrow_{\mathbb{P}} \mathbb{G}_p$ in $\ell^{\infty}(\mathcal{Y})$. $\square$

Proof of Theorem 3. Lemma 19 establishes conditional convergence for the path-oracle multiplier process. Lemma 17 and the conditional multiplier maximal inequality imply

$$\left\| \frac{1}{\sqrt{M}} \sum_{i=1}^M \xi_i \{\widehat{\Psi}_{i,\cdot} - \Psi_{M,\cdot}(O_i)\} \right\|_{\mathcal{Y}} = o_{\mathbb{P}}(1).$$

This proves (9). Continuity of the supremum norm and of the distribution of $\|\mathbb{G}_p\|_{\mathcal{Y}}$ at its $(1 - \alpha)$ -quantile gives consistency of $c_{1-\alpha}^{\xi}$ and proves (10). $\square$

Proposition 5 (Finite multiplier simulation). *Let $c_{1-\alpha}^{\xi, B}$ be the empirical $(1 - \alpha)$ -quantile computed from B independent multiplier draws conditional on the data. If the conditional multiplier-supremum*

distribution is continuous at $c_{1-\alpha}^\xi$, then, as $B \rightarrow \infty$,

$$c_{1-\alpha}^{\xi,B} - c_{1-\alpha}^\xi \rightarrow_{\mathbb{P}} 0.$$

Proof. Conditional on the data, the multiplier supremum draws are independent and identically distributed. The conditional Glivenko–Cantelli theorem gives uniform convergence of their empirical distribution function, and quantile consistency follows at the stated continuity point. $\square$

13 Independent-cluster process inference

The EPA analysis assigns calendar weeks to fitting, weight selection, and CDF evaluation folds and uses one multiplier per week. This section gives the corresponding cluster-level result.

Let $g = 1, \dots, G$ index independent clusters, with observations O_{gj} , $j = 1, \dots, m_g$, and total size $M_G = \sum_{g=1}^G m_g$. Observations within a cluster may be dependent. Regression fitting, weight selection, and CDF evaluation folds are assigned at the cluster level. For an evaluation observation, let $\phi_{gj,y}$ denote the oracle influence function before multiplication by $\sqrt{p_G h^d}$, and define

$$\Psi_{G,g,y}^{\text{cl}} = \sqrt{\frac{G p_G h^d}{M_G}} \sum_{j=1}^{m_g} \phi_{gj,y}. \quad (65)$$

Then

$$\frac{1}{\sqrt{G}} \sum_{g=1}^G \Psi_{G,g,y}^{\text{cl}} = \sqrt{M_G p_G h^d} \left\{ \frac{1}{M_G} \sum_{g=1}^G \sum_{j=1}^{m_g} \phi_{gj,y} \right\}.$$

Proposition 6 (Cluster Gaussian and multiplier processes). *Suppose:*

1. *the clusters are independent, $M_G/G \rightarrow \bar{m} \in (0, \infty)$, and $M_G p_G h^d \rightarrow \infty$;*
2. *fold assignments are made by cluster and the fitted-path replacement results hold for the cluster empirical process;*
3. *the cluster influence classes are pointwise measurable and satisfy a uniform polynomial entropy*

bound;

4. for a cluster envelope F_G^{cl} ,

$$G^{-1} \sum_{g=1}^G E \left[(F_G^{\text{cl}})^2 1\{F_G^{\text{cl}} > \eta \sqrt{G}\} \right] \rightarrow 0$$

for every $\eta > 0$;

5. the covariance kernels

$$\frac{1}{G} \sum_{g=1}^G E \{ \Psi_{G,g,y}^{\text{cl}} \Psi_{G,g,z}^{\text{cl}} \}$$

converge uniformly to a continuous kernel $\Gamma_p^{\text{cl}}(y, z)$, and the associated increment semimetrics are uniformly totally bounded.

Then

$$\sqrt{M_G p_G h^d} \{ \widehat{F}_h^{\text{raw}} - F_{h,0} \} \Rightarrow \mathbb{G}_p^{\text{cl}} \quad \text{in } \ell^\infty(\mathcal{Y}), \quad (66)$$

where $\mathbb{G}_p^{\text{cl}}$ has covariance kernel Γ_p^{cl} .

Let $\widehat{\phi}_{gj,y}$ be the fitted influence functions and let $\xi_1, \dots, \xi_G$ be independent Rademacher multipliers. If the fitted cluster influence functions converge to their oracle counterparts in empirical cluster L_2 , then

$$\mathbb{G}_G^{\text{cl},\xi}(y) = \sqrt{\frac{p_G h^d}{M_G}} \sum_{g=1}^G \xi_g \sum_{j=1}^{m_g} \widehat{\phi}_{gj,y} \quad (67)$$

converges conditionally in probability to $\mathbb{G}_p^{\text{cl}}$ in $\ell^\infty(\mathcal{Y})$. Its supremum quantiles give simultaneous cluster-robust confidence bands.

Proof. After ratio and fitted-path replacement, the leading process is the independent triangular array

$$G^{-1/2} \sum_{g=1}^G \Psi_{G,g,\cdot}^{\text{cl}}.$$

Conditions 3–5 give the entropy, Lindeberg, covariance, and equicontinuity requirements in the

triangular-array functional central limit frameworks of Ziegler (1997) and Arcones (1999). This proves (66).

Conditional on the clusters, the summands in (67) are independent and centered. The cluster product-class maximal inequality gives empirical covariance consistency, and Condition 4 gives conditional finite-dimensional Lindeberg convergence. The conditional entropy bound and empirical increment semimetric yield asymptotic equicontinuity by the argument in Lemma 19. Empirical cluster- L_2 replacement transfers the limit from oracle to fitted cluster influences. $\square$

If $\max_g m_g \leq m_{\max} < \infty$, the observation-level envelope and entropy conditions used in Theorem 3 imply the cluster envelope condition. More general cluster sizes are covered by the displayed Lindeberg condition.

14 Point targets and implementation extensions

14.1 Localization bias and conditional quantiles

Lemma 20 (Uniform localization bias). *Under Conditions 2 and 8,*

$$\sup_{y \in \mathcal{Y}} |F_{h,0}(y \mid x_0) - F_0(y \mid x_0)| = O(h^2).$$

Proof. Let $g_y(x) = q_{0,y}(x)f_X(x)$. Then

$$E(K_h Z_y) = h^d \int K(u) g_y(x_0 + hu) du, \quad \mu_h = h^d \int K(u) f_X(x_0 + hu) du.$$

Uniform second-order Taylor expansions and symmetry of K give

$$\begin{aligned} E(K_h Z_y) &= h^d \{\kappa_1 g_y(x_0) + O(h^2)\}, \\ \mu_h &= h^d \{\kappa_1 f_X(x_0) + O(h^2)\}, \end{aligned}$$

uniformly in y . Taking the ratio proves the result. $\square$

Proof of the point-target statement in Corollary 1. By Lemma 20,

$$\sqrt{N_{M,h}} \|F_{h,0} - F_0\|_{\mathcal{Y}} = O\{\sqrt{N_{M,h}} h^2\} = o(1).$$

The process and multiplier conclusions follow from Theorems 2 and 3 and Slutsky's theorem in $\ell^\infty(\mathcal{Y})$. $\square$

Proposition 7 (Increasing rearrangement). *Suppose $F_0(\cdot \mid x_0)$ is continuously differentiable and strictly increasing on the threshold region, with derivative bounded away from zero and values in $[\eta, 1 - \eta]$. If*

$$r_M(\widehat{F}_h^{\text{raw}} - F_0) \Rightarrow \mathbb{G}_p \quad \text{in } \ell^\infty(\mathcal{Y}), \quad r_M = \sqrt{N_{M,h}},$$

then

$$r_M(\widehat{F}_h^{\text{mon}} - F_0) = r_M(\widehat{F}_h^{\text{raw}} - F_0) + o_{\mathbb{P}}(1) \quad \text{in } \ell^\infty(\mathcal{Y}), \quad (68)$$

and therefore

$$r_M(\widehat{F}_h^{\text{mon}} - F_0) \Rightarrow \mathbb{G}_p \quad \text{in } \ell^\infty(\mathcal{Y}).$$

For $1 \leq r \leq \infty$, rearrangement does not increase the $L_r(v)$ distance to the monotone target.

Proof. The rearrangement contraction and functional limit result at a strictly increasing target follow from Chernozhukov et al. (2010). At such a target, the Hadamard derivative of increasing rearrangement is the identity. Clipping is inactive with probability tending to one on the stated interior probability region. The functional delta method gives (68) and the asserted weak limit. $\square$

Proof of the conditional-quantile statement in Corollary 1. Define the point-target empirical processes

$$\mathbb{Z}_{M,0}^{\text{raw}} = \sqrt{N_{M,h}}(\widehat{F}_h^{\text{raw}} - F_0), \quad \mathbb{Z}_{M,0}^{\text{mon}} = \sqrt{N_{M,h}}(\widehat{F}_h^{\text{mon}} - F_0).$$

By (68),

$$\|\mathbb{Z}_{M,0}^{\text{mon}} - \mathbb{Z}_{M,0}^{\text{raw}}\|_{\mathcal{Y}} = o_{\mathbb{P}}(1).$$

The point-target statement proved above gives

$$\mathbb{Z}_{M,0}^{\text{raw}} \Rightarrow \mathbb{G}_p \quad \text{in } \ell^\infty(\mathcal{Y}).$$

The inverse-CDF map is Hadamard differentiable at F_0 , tangentially to continuous functions, with derivative

$$g \mapsto -\frac{g\{Q_0(\tau \mid x_0)\}}{f_{Y|X}\{Q_0(\tau \mid x_0) \mid x_0\}}.$$

The functional delta method, Theorem 3.9.4 of van der Vaart and Wellner (1996), yields the uniform Bahadur representation

$$\sup_{\tau \in \mathcal{T}} \left| \sqrt{N_{M,h}} \{\widehat{Q}_h(\tau \mid x_0) - Q_0(\tau \mid x_0)\} + \frac{\mathbb{Z}_{M,0}^{\text{raw}}\{Q_0(\tau \mid x_0)\}}{f_{Y|X}\{Q_0(\tau \mid x_0) \mid x_0\}} \right| = o_{\mathbb{P}}(1). \quad (69)$$

Combining (69) with the point-target process limit proves (11)–(12). $\square$

14.2 Estimated verification probabilities

Lemma 21 (Exact propensity drift). *Let $\widehat{\pi}$ be fitted independently of the observation on which $U_y(a, \widehat{\pi})$ is evaluated. Then*

$$E\{U_y(a, \widehat{\pi}) - m_{0,y} \mid W, \widehat{\pi}\} = (a_y - m_{0,y}) \frac{\widehat{\pi} - \pi_M}{\widehat{\pi}}. \quad (70)$$

Proof. Conditional on W and $\widehat{\pi}$,

$$\begin{aligned} E\{U_y(a, \widehat{\pi}) \mid W, \widehat{\pi}\} &= a_y + \frac{1}{\widehat{\pi}} E[R(Z_y - a_y) \mid W] \\ &= a_y + \frac{\pi_M}{\widehat{\pi}} (m_{0,y} - a_y). \end{aligned}$$

Subtracting $m_{0,y}$ gives (70). $\square$

Proposition 8 (Estimated-propensity process extension). *Suppose the conditions of Theorem 2 hold*

for the estimator using π_M . Let $\widehat{\pi}$ be fitted on a separate fold and satisfy, uniformly over $y \in \mathcal{Y}$,

$$\sqrt{N_{M,h}} \left| \frac{1}{\mu_h} E \left[K_h(a_y - m_{0,y}) \frac{\widehat{\pi} - \pi_M}{\widehat{\pi}} \mid \mathcal{T} \right] \right| = o_{\mathbb{P}}(1), \quad (71)$$

$$\frac{p_M h^d}{\mu_h^2} E \left[K_h^2 R(Z_y - a_y)^2 \left(\frac{1}{\widehat{\pi}} - \frac{1}{\pi_M} \right)^2 \mid \mathcal{T} \right] = o_{\mathbb{P}}(1), \quad (72)$$

and suppose the corresponding random class satisfies Condition 5. Then replacing π_M by $\widehat{\pi}$ leaves the limit in Theorem 2 unchanged.

Proof. Decompose

$$U_y(a, \widehat{\pi}) - U_y(a, \pi_M) = R(Z_y - a_y) \left(\frac{1}{\widehat{\pi}} - \frac{1}{\pi_M} \right).$$

Lemma 21 identifies the conditional mean. Condition (71) removes the localized mean at the process scale. After centering, Condition (72) and the entropy bound make the empirical process $o_{\mathbb{P}}(1)$ by the maximal-inequality argument in Lemma 11. The process expansion using π_M therefore remains unchanged. $\square$

Define

$$\begin{aligned} r_{a,M,h}^2 &= \sup_y \frac{1}{\mu_h} E \{ K_h(a_y - m_{0,y})^2 \mid \mathcal{T} \}, \\ r_{\pi,M,h}^2 &= \frac{1}{\mu_h} E \left[K_h \left(\frac{\widehat{\pi} - \pi_M}{\widehat{\pi}} \right)^2 \mid \mathcal{T} \right]. \end{aligned}$$

Cauchy–Schwarz gives

$$\sup_y \left| \frac{1}{\mu_h} E \left[K_h(a_y - m_{0,y}) \frac{\widehat{\pi} - \pi_M}{\widehat{\pi}} \mid \mathcal{T} \right] \right| \leq r_{a,M,h} r_{\pi,M,h}.$$

Thus

$$\sqrt{N_{M,h}} r_{a,M,h} r_{\pi,M,h} = o_{\mathbb{P}}(1)$$

is sufficient for (71). Along an interior limiting path,

$$a_{\lambda,y} - m_{0,y} \rightarrow (1 - \lambda)\{q_{0,y} - m_{0,y}\},$$

so the relevant propensity rate is a localized relative-error rate.

14.3 Grid approximation

Let ν have a bounded density on $\mathcal{Y}$, let

$$\underline{y} = y_{1,L} < \cdots < y_{L,L} = \bar{y},$$

and let $\nu_L = \sum_{j=1}^L w_{j,L} \delta_{y_{j,L}}$ be a positive quadrature rule with mesh Δ_L .

Proposition 9 (Grid-to-continuum transfer). *Under Conditions 1 and 2, suppose, uniformly in the local region,*

$$|A_y D_y - A_z D_z| + |D_y^2 - D_z^2| \leq C_\beta |y - z|^\beta$$

for some $\beta > 0$ and $C_\beta < \infty$. Suppose the quadrature rule has order β . Let $(G_{M,h}, H_{M,h})$ use ν and $(G_{M,h,L}, H_{M,h,L})$ use ν_L . Then

$$|G_{M,h,L} - G_{M,h}| + |H_{M,h,L} - H_{M,h}| \leq C \Delta_L^\beta E \nu_{M,h}.$$

After multiplication by p_M , the right side is $O(\Delta_L^\beta)$. If $\inf_M \mathcal{H}_{M,h} > 0$,

$$|\lambda_{h,L}^{\text{or}} - \lambda_h^{\text{or}}| = O(\Delta_L^\beta).$$

Proof. The quadrature error for a uniformly β -Hölder function is $O(\Delta_L^\beta)$. Apply this result pointwise to $A_y D_y$ and D_y^2 , multiply by $\nu_{M,h}$, and take expectation. Since $p_M E \nu_{M,h} = O(1)$, the normalized error is $O(\Delta_L^\beta)$. Projection is nonexpansive and the ratio map is Lipschitz when $\mathcal{H}$ is bounded below. $\square$

Proposition 10 (Growing-grid approximation of process suprema). *Let $\mathcal{Y}_{L_M} \subset \mathcal{Y}$ be deterministic grids with mesh $\Delta_{L_M} \rightarrow 0$. Under Theorems 2 and 3,*

$$\left| \|Z_M\|_{\mathcal{Y}} - \max_{y \in \mathcal{Y}_{L_M}} |Z_M(y)| \right| = o_{\mathbb{P}}(1),$$

$$\left| \|G_M^\xi\|_{\mathcal{Y}} - \max_{y \in \mathcal{Y}_{L_M}} |G_M^\xi(y)| \right| = o_{\mathbb{P}}(1)$$

conditionally in probability for the multiplier process. If the reported CDF is extended by piecewise-linear interpolation and $F_{h,0}$ has a uniformly bounded derivative, the deterministic interpolation error is negligible whenever

$$\sqrt{N_{M,h}} \Delta_{L_M} \rightarrow 0.$$

Proof. The process and multiplier proofs establish asymptotic equicontinuity under the totally bounded semimetric ρ . The maximal ρ -distance from any point in $\mathcal{Y}$ to the grid tends to zero, and each supremum difference is bounded by the corresponding modulus of continuity. Piecewise-linear interpolation of the target has error at most $C\Delta_{L_M}$, while stochastic interpolation is controlled by the same process modulus. $\square$

14.4 Local-linear equivalent-kernel extension

The local-constant construction extends to local-polynomial distribution estimation and related boundary-adaptive conditional-density procedures through equivalent-kernel representations; see Cattaneo et al. (2024b) and Cattaneo et al. (2024a). We give the interior local-linear version for the selective-verification model.

Let

$$r_h(X) = r\{(X - x_0)/h\}, \quad r(u) = (1, u^\top)^\top,$$

and define

$$\mathbf{S}_h = E\{K_h(X - x_0)r_h(X)r_h(X)^\top\}, \quad (73)$$

$$\gamma_{h,y} = E\{K_h(X - x_0)r_h(X)Z_y\}, \quad \beta_{h,y} = \mathbf{S}_h^{-1}\gamma_{h,y}. \quad (74)$$

The local-linear regularized target is

$$F_{h,0}^{\text{LL}}(y \mid x_0) = e_1^\top \beta_{h,y}, \quad e_1 = (1, 0, \dots, 0)^\top \in \mathbb{R}^{d+1}. \quad (75)$$

For a path a fixed conditional on data outside the evaluation sample, let

$$\begin{aligned} \widehat{\mathbf{S}}_h &= \mathbb{P}_M\{K_h r_h r_h^\top\}, \\ \widehat{\gamma}_{h,y}^a &= \mathbb{P}_M\{K_h r_h U_y(a)\}, \\ \widehat{\beta}_{h,y}^a &= \widehat{\mathbf{S}}_h^{-1} \widehat{\gamma}_{h,y}^a, \\ \widehat{F}_h^{\text{LL},a}(y \mid x_0) &= e_1^\top \widehat{\beta}_{h,y}^a. \end{aligned}$$

The sample-split estimator applies this construction foldwise and combines the evaluation folds as in (5). Define the population equivalent weight

$$L_h(X; x_0) = e_1^\top \mathbf{S}_h^{-1} r_h(X) K_h(X - x_0). \quad (76)$$

Proposition 11 (Local-linear efficiency, transfer, and process). *Under Condition 1, fix M and h such that $\mathbf{S}_h$ is nonsingular and*

$$E\{L_h^2(X; x_0)/\pi_M(W)\} < \infty.$$

For a finite threshold vector, the canonical-gradient components are

$$\phi_{M,h,y}^{\text{LL,eff}}(O) = L_h(X; x_0) \left[m_{0,y}(W) + \frac{R}{\pi_M(W)} \{Z_y - m_{0,y}(W)\} - r_h(X)^\top \beta_{h,y} \right]. \quad (77)$$

For a generic augmentation, set

$$\phi_{M,h,y}^{\text{LL},a}(O) = L_h(X; x_0) \{U_y(a) - r_h(X)^\top \beta_{h,y}\}, \quad (78)$$

and

$$\Omega_{M,h}^{\text{LL},a}(y, z) = h^d E \{ \phi_{M,h,y}^{\text{LL},a} \phi_{M,h,z}^{\text{LL},a} \}.$$

Under a known verification law, $\phi^{\text{LL},a}$ is an influence function and

$$\Omega_{M,h}^{\text{LL},a}(y, z) - \Omega_{M,h}^{\text{LL},m_0}(y, z) = h^d E \left[L_h^2(X; x_0) \left\{ \frac{1}{\pi_M(W)} - 1 \right\} \delta_y^a(W) \delta_z^a(W) \right]. \quad (79)$$

The difference is positive semidefinite over every finite threshold collection, and $\phi^{\text{LL},m_0} = \phi^{\text{LL,eff}}$ remains canonical when the verification law is unrestricted.

Suppose next that Condition 2 holds and that the sample-separation, curvature, and candidate-path clauses of Condition 3 hold for the local-linear equivalent-kernel metric. Suppose that

$$h^{-d} \mathbf{S}_h \rightarrow f_X(x_0) \mathbf{S}_K, \quad \mathbf{S}_K = \int K(u) r(u) r(u)^\top du,$$

where $\mathbf{S}_K$ is positive definite. Assume the equivalent-kernel versions of Conditions 4, 5, 6, and 7 hold for L_h , the fitted paths, and their threshold increments. Then

$$\sqrt{N_{M,h}} \{ \widehat{F}_h^{\text{LL}}(a_M; \cdot \mid x_0) - F_{h,0}^{\text{LL}}(\cdot \mid x_0) \} \Rightarrow \mathbb{G}_p^{\text{LL}} \quad \text{in } \ell^\infty(\mathcal{Y}). \quad (80)$$

The transfer criterion, observable gain representation, coefficient estimator, path-oracle equivalence,

and multiplier process remain valid after replacing $v_{M,h}$ by

$$v_{M,h}^{\text{LL}}(W) = h^d L_h^2(X; x_0) \left\{ \frac{1}{\pi_M(W)} - 1 \right\}. \quad (81)$$

If

$$\ell_K(u) = e_1^\top \mathbf{S}_K^{-1} r(u) K(u), \quad \kappa_K^{\text{LL}} = \int \ell_K^2(u) du,$$

then the covariance kernel in Theorem 2 has κ_K replaced by κ_K^{LL} . Under Condition 8,

$$\sup_{y \in \mathcal{Y}} |F_{h,0}^{\text{LL}}(y | x_0) - F_0(y | x_0)| = O(h^2),$$

so the point-target and quantile conclusions in Corollary 1 apply to the local-linear estimator.

Proof. The augmentation identity gives

$$E\{K_h r_h U_y(a)\} = E(K_h r_h Z_y) = \gamma_{h,y},$$

so $F_{h,0}^{\text{LL}}$ is independent of the augmentation. The parameter $\beta_{h,y}$ solves

$$E[K_h r_h \{Z_y - r_h^\top \beta_{h,y}\}] = 0.$$

Differentiating this moment equation along a full-data parametric submodel with score s gives

$$\dot{\beta}_{h,y} = \mathbf{S}_h^{-1} E[K_h r_h \{Z_y - r_h^\top \beta_{h,y}\} s].$$

Hence the full-data gradient for $F_{h,0}^{\text{LL}}(y | x_0)$ is

$$L_h(X; x_0) \{Z_y - r_h(X)^\top \beta_{h,y}\}.$$

Applying the observed-data tangent decomposition in Section 4 replaces this gradient by its conditional mean given W plus the inverse-probability residual, which is exactly (77). The same

propensity-tangent calculation as in Theorem 1 proves canonicity under an unrestricted verification law.

The sample and population normal equations give the exact identity

$$\widehat{\beta}_{h,y}^a - \beta_{h,y} = \widehat{\mathbf{S}}_h^{-1} (\mathbb{P}_M - P) [K_h r_h \{U_y(a) - r_h^\top \beta_{h,y}\}]. \quad (82)$$

Every entry of $K_h r_h r_h^\top$ is bounded and supported on a set of probability order h^d . Therefore

$$\|\widehat{\mathbf{S}}_h - \mathbf{S}_h\| = O_{\mathbb{P}}\{(h^d/M)^{1/2}\}, \quad \|\mathbf{S}_h^{-1}(\widehat{\mathbf{S}}_h - \mathbf{S}_h)\| = O_{\mathbb{P}}\{(Mh^d)^{-1/2}\}. \quad (83)$$

The matrix inverse expansion is

$$\widehat{\mathbf{S}}_h^{-1} = \mathbf{S}_h^{-1} - \mathbf{S}_h^{-1}(\widehat{\mathbf{S}}_h - \mathbf{S}_h)\mathbf{S}_h^{-1} + \mathcal{R}_{M,h}^{\mathbf{S}}, \quad (84)$$

with

$$\|\mathcal{R}_{M,h}^{\mathbf{S}}\| = O_{\mathbb{P}}\{h^{-d}(Mh^d)^{-1}\}.$$

The empirical score in (82) is

$$O_{\mathbb{P}}\{(h^d/(Mp_M))^{1/2}\}$$

in each fixed direction and uniformly over thresholds under Condition 5. Thus

$$\|\widehat{\mathbf{S}}_h^{-1} - \mathbf{S}_h^{-1}\| O_{\mathbb{P}}\{(h^d/(Mp_M))^{1/2}\} = O_{\mathbb{P}}\left\{N_{M,h}^{-1/2}(Mh^d)^{-1/2}\right\} = o_{\mathbb{P}}(N_{M,h}^{-1/2}).$$

Equation (78) is therefore the first-order influence function.

The difference between $\phi^{\text{LL},a}$ and ϕ^{LL,m_0} is

$$L_h(X; x_0) \delta_y^a(W) \left(1 - \frac{R}{\pi_M(W)}\right).$$

It is orthogonal to the known-design tangent space. Lemma 2 gives (79), and its finite-dimensional

quadratic forms are nonnegative. Integration over ν produces the quadratic transfer criterion with weight (81); Lemma 5 gives the observable alignment representation. The gain-estimation, coefficient, path-replacement, and multiplier arguments then apply foldwise with the equivalent weight.

Finally,

$$h^{-d}\mathbf{S}_h \rightarrow f_X(x_0)\mathbf{S}_K$$

implies

$$L_h(x_0 + hu; x_0) = h^{-d} f_X(x_0)^{-1} \ell_K(u) + o(h^{-d})$$

uniformly on the support of K . The change-of-variables argument in Lemma 14 replaces κ_K by κ_K^{LL} . The envelope, increment, and entropy calculations retain the orders established in Sections 9–12, proving the process and multiplier limits. A uniform second-order expansion of $q_{0,y}(x)$ around x_0 , combined with the local-linear moment equations, gives the $O(h^2)$ bias. $\square$

14.5 Adaptive borrowing with estimated verification probabilities

Proposition 8 treats propensity replacement during process evaluation for a fixed cross-fitted augmentation. The adaptive procedure also uses the verification probabilities in the gain criterion and therefore requires control of the selected coefficient.

For evaluation fold k , enlarge $\mathcal{T}_k$ to include all data used to fit $\widehat{q}_k$, $\widehat{m}_k$, and $\widehat{\pi}_k$. The propensity regression is fitted outside the gain and evaluation folds. For a verification function r , let

$$(\widehat{\mathcal{G}}_{h,k}(r), \widehat{\mathcal{H}}_{h,k}(r), \widehat{\lambda}_{h,k}(r))$$

denote the gain quantities computed with r in both the transfer weight and the inverse-probability residual. We compare $r = \pi_M$ and $r = \widehat{\pi}_k$, holding the threshold regressions and selection-fold

observations fixed. Write

$$\widehat{a}_{k,y}(\lambda) = \widehat{q}_{k,y} + \lambda \widehat{D}_{k,y}, \quad \widehat{D}_k = \widehat{m}_k - \widehat{q}_k,$$

and set

$$\widehat{a}_k[r] = \widehat{a}_k\{\widehat{\lambda}_{h,k}(r)\}.$$

Proposition 12 (Adaptive process with estimated verification probabilities). *Suppose the conditions of Theorems 2 and 3 hold for the known-design procedure. Assume, uniformly over the three rotations,*

$$|\widehat{\mathcal{G}}_{h,k}(\widehat{\pi}_k) - \widehat{\mathcal{G}}_{h,k}(\pi_M)| + |\widehat{\mathcal{H}}_{h,k}(\widehat{\pi}_k) - \widehat{\mathcal{H}}_{h,k}(\pi_M)| = o_{\mathbb{P}}(1). \quad (85)$$

Assume Conditions (71)–(72) hold uniformly over

$$\{\widehat{a}_k(\lambda) : 0 \leq \lambda \leq 1\}.$$

For the coefficient direction, assume

$$\sqrt{N_{M,h}} \sup_{y \in \mathcal{Y}} \left| \frac{1}{\mu_h} E \left[K_h \widehat{D}_{k,y}(W) \frac{\widehat{\pi}_k(W) - \pi_M(W)}{\widehat{\pi}_k(W)} \mid \mathcal{T}_k \right] \right| = o_{\mathbb{P}}(1), \quad (86)$$

$$\sup_{y \in \mathcal{Y}} \frac{p_M h^d}{\mu_h^2} E \left[K_h^2 R \widehat{D}_{k,y}^2(W) \left\{ \frac{1}{\widehat{\pi}_k(W)} - \frac{1}{\pi_M(W)} \right\}^2 \mid \mathcal{T}_k \right] = o_{\mathbb{P}}(1). \quad (87)$$

Assume the propensity-induced evaluation, gain, and direction classes satisfy Condition 5. Then

$$\sqrt{N_{M,h}} \left\| \widehat{F}_h[\widehat{\pi}] - \widehat{F}_h[\pi_M] \right\|_{\mathcal{Y}} = o_{\mathbb{P}}(1). \quad (88)$$

The Gaussian-process and multiplier limits coincide with those of the known-design estimator.

Proof. For fold k , decompose

$$\widehat{F}_{h,k}[\widehat{\pi}_k] - \widehat{F}_{h,k}[\pi_M] = \{\widehat{F}_{h,k}(\widehat{a}_k[\widehat{\pi}_k], \widehat{\pi}_k) - \widehat{F}_{h,k}(\widehat{a}_k[\pi_M], \widehat{\pi}_k)\} \quad (89)$$

$$+ \{\widehat{F}_{h,k}(\widehat{a}_k[\pi_M], \widehat{\pi}_k) - \widehat{F}_{h,k}(\widehat{a}_k[\pi_M], \pi_M)\}. \quad (90)$$

For (90), Lemma 21, Conditions (71)–(72), and the conditional maximal inequality give a uniform $o_{\mathbb{P}}(N_{M,h}^{-1/2})$ bound.

For (89),

$$U_y(\widehat{a}_k[\widehat{\pi}_k], \widehat{\pi}_k) - U_y(\widehat{a}_k[\pi_M], \widehat{\pi}_k) = d_{\lambda,k} \widehat{D}_{k,y}(W) \left(1 - \frac{R}{\widehat{\pi}_k(W)}\right), \quad (91)$$

where

$$d_{\lambda,k} = \widehat{\lambda}_{h,k}(\widehat{\pi}_k) - \widehat{\lambda}_{h,k}(\pi_M).$$

Its localized conditional mean is

$$\frac{1}{\mu_h} E \left[K_h \widehat{D}_{k,y}(W) \left\{ 1 - \frac{\pi_M(W)}{\widehat{\pi}_k(W)} \right\} \mid \mathcal{T}_k \right]. \quad (92)$$

Condition (86) makes the scaled mean contribution negligible because $|d_{\lambda,k}| \leq 1$.

For the centered component, use

$$1 - \frac{R}{\widehat{\pi}_k} = \left(1 - \frac{R}{\pi_M}\right) + R \left(\frac{1}{\pi_M} - \frac{1}{\widehat{\pi}_k}\right).$$

The first term has localized squared radius

$$\sup_y \frac{p_M h^d}{\mu_h^2} E \left[K_h^2 \left(\frac{1}{\pi_M} - 1\right) \widehat{D}_{k,y}^2 \mid \mathcal{T}_k \right],$$

and Condition (87) makes the centered second term negligible. If $\mathcal{H}_{h,k} \rightarrow H_{\infty,k} > 0$, consistency

of the known-design gain quantities, (85), and local Lipschitz continuity of the projected ratio give

$$d_{\lambda,k} = o_{\mathbb{P}}(1),$$

while the first radius is $O_{\mathbb{P}}(1)$. If $\mathcal{H}_{h,k} \rightarrow 0$, Condition 6 makes this radius tend to zero, while the coefficient difference remains bounded by one. The conditional maximal inequality and the ratio linearization therefore give

$$\sqrt{N_{M,h}} \left\| \widehat{F}_{h,k}(\widehat{a}_k[\widehat{\pi}_k], \widehat{\pi}_k) - \widehat{F}_{h,k}(\widehat{a}_k[\pi_M], \widehat{\pi}_k) \right\|_{\mathcal{Y}} = o_{\mathbb{P}}(1).$$

Summing over the evaluation folds proves (88).

For the multiplier process, the estimated-propensity minus known-design influence difference has the same mean-plus-centered decomposition. The drift conditions remove its empirical centering at the $N_{M,h}^{-1/2}$ scale, and the two variance conditions give a uniform empirical L_2 norm of order $o_{\mathbb{P}}(1)$. Lemma 17 and the conditional multiplier maximal inequality then transfer the known-design multiplier limit. $\square$

A lower-level route uses local uniform relative consistency. Let C_K satisfy $K(u) = 0$ for $\|u\| > C_K$, and define

$$\epsilon_{\pi,M,h} = \text{ess sup}_{\|X-x_0\| \leq C_K h} \left| \frac{\widehat{\pi}_k(W)}{\pi_M(W)} - 1 \right|.$$

If $\epsilon_{\pi,M,h} = o_{\mathbb{P}}(1)$, $\widehat{\pi}_k/\pi_M$ is bounded above and away from zero on the local region, and the candidate paths are bounded, then direct expansion of the normalized gain integrands gives

$$|\widehat{\mathcal{G}}_{h,k}(\widehat{\pi}_k) - \widehat{\mathcal{G}}_{h,k}(\pi_M)| + |\widehat{\mathcal{H}}_{h,k}(\widehat{\pi}_k) - \widehat{\mathcal{H}}_{h,k}(\pi_M)| = O_{\mathbb{P}}(\epsilon_{\pi,M,h}). \quad (93)$$

Define

$$r_{D,M,h}^2 = \sup_y \frac{1}{\mu_h} E\{K_h \widehat{D}_{k,y}^2 \mid \mathcal{T}_k\}.$$

The same relative-consistency condition makes the left side of (87) $O_{\mathbb{P}}(\epsilon_{\pi,M,h}^2)$. Moreover,

$$\sqrt{N_{M,h}} r_{D,M,h} r_{\pi,M,h} = o_{\mathbb{P}}(1)$$

controls (86), while

$$\sqrt{N_{M,h}} r_{a,M,h} r_{\pi,M,h} = o_{\mathbb{P}}(1)$$

controls the evaluation drift uniformly over the fitted segment. Together with (93), these conditions yield the adaptive process and multiplier conclusions.

14.6 Estimating the overall reference fraction

The parameterization

$$p_M = E\{\pi_M(W)\}, \quad e_M(W) = \pi_M(W)/p_M$$

is used to state the triangular rate and covariance limit. The adaptive coefficient does not require p_M , because multiplying $G_{M,h}$ and $H_{M,h}$ by the same positive scalar leaves their ratio unchanged. For process scaling under an estimated verification law, p_M can be estimated directly from the observed verification indicators.

Proposition 13 (Reference-fraction estimation). *Let*

$$\widehat{p}_M = \frac{1}{M} \sum_{i=1}^M R_i.$$

Under Conditions 1 and 2,

$$\frac{\widehat{p}_M}{p_M} - 1 = O_{\mathbb{P}}\{(Mp_M)^{-1/2}\} = o_{\mathbb{P}}(1). \quad (94)$$

Consequently,

$$\sqrt{M\widehat{p}_M h^d} = \sqrt{Mp_M h^d} \{1 + o_{\mathbb{P}}(1)\},$$

and replacing p_M by $\widehat{p}_M$ in process normalization, multiplier normalization, and the reported local-information rate index leaves the limiting laws unchanged.

If

$$\widehat{p}_M^\pi = \frac{1}{M} \sum_{i=1}^M \widehat{\pi}(W_i)$$

is formed from cross-fitted propensity predictions and

$$\frac{1}{M} \sum_{i=1}^M \frac{|\widehat{\pi}(W_i) - \pi_M(W_i)|}{p_M} = o_{\mathbb{P}}(1),$$

then $\widehat{p}_M^\pi / p_M \rightarrow_{\mathbb{P}} 1$ and the same conclusion holds.

Proof. Since $E(R_i) = p_M$,

$$\text{Var}(\widehat{p}_M) = \frac{1}{M} \text{Var}(R_i) \leq \frac{p_M}{M}.$$

Chebyshev's inequality gives (94). The condition $M p_M h^d \rightarrow \infty$, together with $h^d \leq 1$ eventually, implies $M p_M \rightarrow \infty$. The square-root expansion follows from differentiability at one. For $\widehat{p}_M^\pi$,

$$\left| \frac{\widehat{p}_M^\pi}{p_M} - 1 \right| \leq \left| \frac{M^{-1} \sum_i \pi_M(W_i)}{p_M} - 1 \right| + \frac{1}{M} \sum_i \frac{|\widehat{\pi}(W_i) - \pi_M(W_i)|}{p_M}.$$

The first term is $O_{\mathbb{P}}(M^{-1/2})$ because $e_M = \pi_M / p_M$ is uniformly bounded; the second tends to zero by assumption. $\square$

15 Additional numerical and empirical results

All results in this section were generated with the implementation version recorded in the result registry. The registry contains 95 point/risk simulation objects, 24 multiplier-band objects, 115 AI downstream objects, 6 AI band objects, and two application objects (242 objects total), each with an exact path and SHA-256 hash.

15.1 Simulation inventory

The simulation manifest contains 119 configurations spanning oracle mechanisms, a five-cell vanishing-curvature path, estimated regression paths, selective verification, scarce-reference scaling, nuisance and distributional perturbations, multiplier bands, estimated propensities, discrete outcomes, and two-dimensional localization. Every replication uses the same generated data, verification mask, fold assignment, and threshold grid across methods. Primary modules use 2,000 replications, secondary modules use 1,000, and two-dimensional modules use 500. The multiplier-band module uses 4,999 Rademacher draws per replication. L’Ecuyer–CMRG streams assign deterministic, scenario-specific data streams. Multiplier seeds are deterministic functions of replication and method, are recorded in every object, and are shared across band scenarios to provide common multiplier draws.

15.2 Threshold-level CDF evaluation

The scalar result objects retain ISE, tail loss, quantile loss, borrowing, and coverage summaries but not every threshold-level estimate. Before producing the new curves, we selected three existing settings to represent aligned borrowing under 10% random verification, a moderate surrogate under 5% selective verification, and local reversal under 10% random verification. The choice was based on these scientific roles rather than the appearance of the curves. Each setting was recomputed over its first 500 deterministic random-number streams without changing its data-generating mechanism, learner, bandwidth, folds, or threshold grid. The artifact stores the truth, mean and median rearranged CDF, pointwise bias, pointwise RMSE, central estimate quantiles, the first-stream diagnostic bands, all seeds, and source hashes. All twelve integrity checks pass.

The application panels use the first reference sample at the following targets: 50% relative humidity for EPA, 75 minutes for VitalDB, and 10% random review at $x_0 = 0$ for the AI study. The EPA and AI Adaptive bands contain their complete-data or population benchmarks. The VitalDB band does not, and its fixed-mask ISE ordering differs from the repeated-mask mean ordering. These panels remain diagnostic examples rather than empirical coverage estimates.

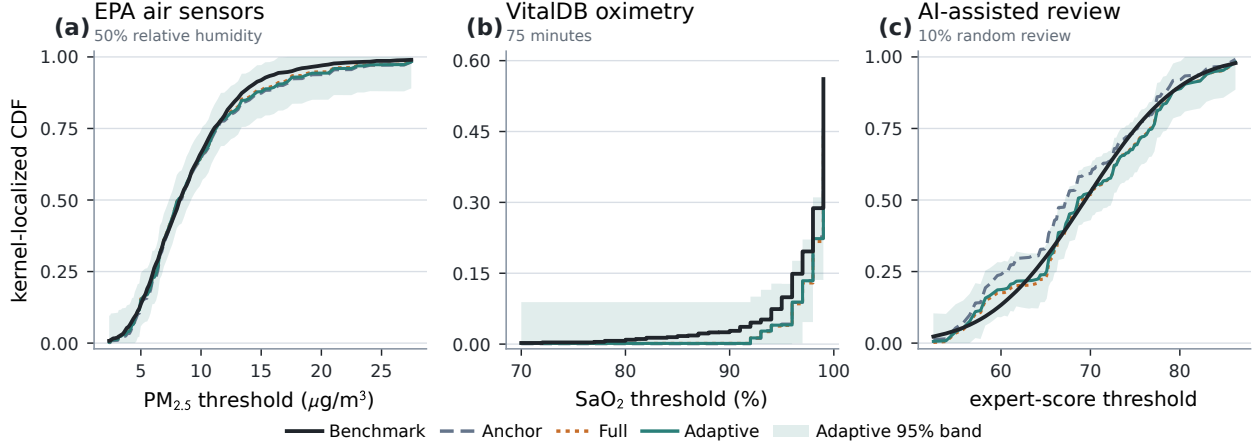


Figure S1: Conditional-CDF estimates from one fixed reference sample in each study. Each panel uses a 10% random-reference sample at a target fixed before the curves were produced. Black denotes $F_{h,\text{emp}}$ for EPA and VitalDB and the frozen $F_{h,0}$ for the AI benchmark; slate dashed, orange dotted, and teal solid curves denote Anchor, Full, and Adaptive. The teal ribbon is the Adaptive 95% simultaneous band. The EPA and AI bands contain their benchmarks; the VitalDB band does not.

15.3 Oracle mechanisms

Supplementary Table S1 reports the oracle geometry. The sign of $2G_h - H_h$ matches the realized anchor-minus-Full ISE direction in all 18 cells. M4 now has separate clipped and raw paths. The clipped paths have positive transfer signals, while the raw paths have negative signals; both versions agree with their realized risk directions. M5–M6 have $\lambda_h^{\text{or}} = 0$, mean estimated coefficients below 10^{-4} , and harmful full borrowing.

Oracle $m_0(X, S)$ reduces mean ISE relative to the X -anchor by 25.3–28.0% in M1–M2. In M2, the conservative candidate improves on the anchor but does not attain the oracle- m risk, illustrating the positive-semidefinite augmentation excess directly.

15.4 Candidate regression methods and local failure

Supplementary Table S2 compares the primary location–scale learner with misspecified parametric and nearest-neighbor candidates. At moderate surrogate noise, Adaptive gains are 23.7%, 25.3%, and 22.4% for the primary, L2, and L3 learners. Across the strong-to-weak surrogate sequence, primary gains decline from 48.5% to 7.6%; the analogous L2 range is 49.8–11.0% and the L3 range

is 37.0–8.0%.

The local-reversal scenario N5 is the main failure audit. Adaptive reduces mean borrowing to 0.0017 and negative-transfer frequency to 1.5%, compared with 82.3% for Full. Its mean ISE remains 0.066% above the anchor because a small number of loss realizations dominate the paired mean. This result shows that Adaptive greatly reduces the risk increase caused by local reversal, but does not establish uniform or finite-sample dominance.

15.5 Distributional perturbations and localization dimension

Adaptive gains are 15.6%, 17.8%, and 24.7% under standardized t_3 , standardized χ_3^2 , and Gaussian-mixture errors, compared with 22.0% under Gaussian errors. For the ten-level outcome, category-scale oracle truth produces a 22.5% Adaptive gain and a 25.9% Full gain. The old apparent loss was caused by evaluating discrete outcomes against a continuous-scale truth engine.

With $X = (X_1, X_2)$, a product Epanechnikov kernel, and $h = M^{-1/6}$, Adaptive gains range from 20.9% to 24.4% over $M \in \{2,000, 5,000\}$ and $p_M \in \{0.10, 0.20\}$.

15.6 Bandwidth, curvature, and estimated propensities

The bandwidth and denominator-floor checks preserve the primary risk ordering. The numerical analyses use a denominator floor of 10^{-8} . We vary the candidate-path scale over $s \in \{1, 0.5, 0.25, 0.125, 0\}$, so normalized curvature follows s^2 , the effective correction decreases from 0.899 to zero, and Adaptive-minus-path-oracle regret decreases from 6.01×10^{-5} to zero.

Eight exploratory configurations estimate verification probabilities on the fitting fold. Adaptive gains range from 22.9–45.4%, while mean borrowing ranges from 0.408 to 0.600. These results demonstrate sensitivity of the selected path to propensity estimation; inferential validity still requires the local product-rate condition in Section 14.5.

15.7 Scarcity and multiplier bands

Supplementary Table S3 reports scaling over four sample sizes and four reference fractions. For $Mp_M h \geq 30$, scaled ISE lies in 0.319–0.358 for the anchor and 0.241–0.286 for Adaptive. The rescaling is stable over the grid, while selective placement changes the covariance constant at the same nominal reference fraction.

Supplementary Table S4 reports the 24 multiplier cells. At the lowest nominal rate index, the selective-verification cells give Adaptive coverage 0.910–0.935 and anchor coverage 0.908–0.932. The minimum Adaptive estimate has Monte Carlo standard error 0.0090 and a 95% Wilson interval of 0.8907–0.9262. Adaptive coverage over all cells ranges from 0.910 to 0.953. Mean coverage under selective verification increases from 0.9220 at $p_M = 0.05$ to 0.9435 at $p_M = 0.20$. Aligned surrogates reduce integrated width to 0.760–0.981 of the anchor width; locally reversed paths produce ratios within 1.000–1.003, eliminating nearly all width gain. The minimum role-fold verified support averages about five in the lowest selective cells. Table S5 shows pooled raw coverage of 0.911 for counts 4–5 and 0.945 for counts of at least 10.

A pre-specified seven-cell sensitivity used the same data streams and 4,999 multiplier draws. Increasing rearrangement of the endpoints raised coverage in several low-information cells but did not restore the nominal level uniformly. A pointwise-studentized supremum band produced coverage 0.358–0.633 and was 13–24% narrower than the unstudentized band. It failed the selection rule, so the raw-process band remains primary.

15.8 Application configurations

The EPA analysis uses $X = (RH - 50)/20$, $S = \log(1 + \text{PurpleAir})$, and $Y = \log(1 + \text{reference PM}_{2.5})$. Targets $x_0 \in \{-1, 0, 1\}$ correspond to 30%, 50%, and 70% relative humidity; the Epanechnikov bandwidth is $h = 0.5$, and the 101 thresholds span the empirical 0.01–0.99 quantiles of transformed Y .

The VitalDB analysis uses $X = \log(1 + \text{elapsed minutes})$, $S = \text{SpO}_2$, and $Y = \text{SaO}_2$. Targets are 30, 75, and 180 minutes, the Epanechnikov bandwidth is $h = 0.55$, and the threshold grid is 70:99.

The AI analysis uses standardized complexity X , automated score S , and the mean of three expert ratings as Y , with $x_0 = 0$, $h = 0.3$, and 101 thresholds defined by point-target conditional quantile levels 0.02–0.98. For confidence-triggered designs, the broadly observed vector is $W = (X, S, C)$, where model confidence C enters the known verification probability; the augmentation candidates use (X, S) . The risk benchmark is the frozen population $F_{h,0}$.

15.9 EPA details

The EPA source archive is verified by SHA-256 before extraction. The uncorrected PurpleAir records are aggregated to 2,814 site-days, and calendar weeks are assigned to regression fitting, weight selection, and CDF evaluation. At the 10% random budget, Adaptive gains are 52.2–60.3%, with paired intervals excluding zero. A single weekly cluster-multiplier diagnostic gives 19–30% narrower Adaptive bands and complete-data containment at all three humidity targets; it is not a repeated-coverage result.

15.10 VitalDB details

The VitalDB acquisition and hash-selection rules produce 8,476 eligible pairs and 3,178 subject-level observations. At the 10% random budget, Adaptive gains are 11.8%, 11.2%, and 2.95%. Within-grid information multipliers are 1.109, 1.102, and 1.027 at 30, 75, and 180 minutes. Full has lower mean ISE, but its paired advantage is conclusive only at 30 minutes. Adaptive reduces observed negative-transfer frequency relative to Full at all targets; the paired interval excludes zero at 75 and 180 minutes.

15.11 AI-assisted evaluation details

The corrected reference uses fixed expert biases $(-1, 0, 1)$ while retaining the same dossier text and 5,997 valid model responses. On Corpus A, bias, MAE, and RMSE are 0.350, 2.415, and 3.058; Pearson and Spearman correlations are 0.946 and 0.942.

The primary evaluator is `deepseek-v4-pro`; `deepseek-v4-flash` is used for the model-sensitivity analysis. Both use the frozen dossier text and rubric. A canonical call manifest links each corpus row to its request hash and cached response-file checksum; the model outputs themselves were fixed before any reference mask was drawn. A companion cache manifest checksums all retained primary and sensitivity responses.

Under random review, Adaptive gains are 53.9–59.9%. For fractions up to 0.10, the anchor needs 2.28–2.50 times the actual fraction to match Adaptive risk within the evaluated grid. Under local under-review and local reversal, Adaptive borrowing collapses toward zero and limits mean loss to at most 0.39%, whereas Full remains harmful. Representative AI multiplier cells have Adaptive coverage 0.940–0.974 and narrower bands than the anchor.

15.12 Complete simulation tables

Table S1: Oracle mechanism results. The realized gain is anchor ISE minus full-borrowing ISE.

Scenario	G_h	H_h	$2G_h - H_h$	λ_h^{or}	$E(\widehat{\lambda}_h)$	Realized gain
M1-R1	0.575	0.575	0.575	1.000	0.900	0.001
M1-R2	0.861	0.861	0.861	1.000	0.852	0.001
M2-R1	0.287	0.144	0.431	1.000	0.998	0.001
M2-R2	0.431	0.215	0.646	1.000	0.984	0.001
M3-R1	0.750	0.998	0.501	0.751	0.734	0.001
M3-R2	1.131	1.515	0.746	0.746	0.721	0.001
M4-R1	0.867	1.445	0.290	0.600	0.595	0.001
M4-R1-unclip	1.437	3.593	-0.719	0.400	0.399	-0.001
M4-R2	1.321	2.239	0.403	0.590	0.582	0.001
M4-R2-unclip	2.153	5.381	-1.076	0.400	0.391	-0.002
M5-R1	-0.269	0.130	-0.667	0.000	0.000	-0.001
M5-R2	-0.404	0.196	-1.005	0.000	0.000	-0.002
M6-R1	-0.269	0.130	-0.667	0.000	0.000	-0.001
M6-R2	-0.404	0.196	-1.005	0.000	0.000	-0.002
M7-R1	0.507	0.532	0.481	0.952	0.873	0.001
M7-R2	0.733	0.779	0.687	0.941	0.837	0.001
M8-R1	0.567	0.687	0.448	0.826	0.794	0.001
M8-R2	0.836	1.025	0.646	0.815	0.742	0.001

Table S2: Feasible estimator results. Relative gain is the percentage reduction in mean ISE from the X-anchor.

Scenario	ISE anchor	ISE Adaptive	ISE Full	Rel. gain	$\widehat{\lambda}_h$	NT Adaptive	NT Full
F1-p0.05	0.0099	0.0075	0.0069	23.4	0.693	0.332	0.346
F1-p0.10	0.0048	0.0036	0.0035	24.0	0.784	0.339	0.360
F1-p0.20	0.0024	0.0019	0.0018	21.5	0.850	0.362	0.383
F2-L2-t16	0.0049	0.0044	0.0043	11.0	0.699	0.388	0.419
F2-L2-t4	0.0050	0.0025	0.0023	49.8	0.833	0.219	0.223
F2-L2-t8	0.0047	0.0035	0.0034	25.3	0.781	0.349	0.353
F2-L3-t16	0.0049	0.0045	0.0044	8.0	0.730	0.381	0.396
F2-L3-t4	0.0049	0.0031	0.0030	37.0	0.947	0.207	0.213
F2-L3-t8	0.0050	0.0039	0.0038	22.4	0.880	0.313	0.319
F2-t16	0.0047	0.0044	0.0043	7.6	0.683	0.429	0.443
F2-t4	0.0049	0.0025	0.0023	48.5	0.835	0.203	0.222
F2-t8	0.0047	0.0036	0.0035	23.7	0.787	0.337	0.370
F3-M1000	0.0082	0.0066	0.0063	19.1	0.713	0.379	0.391
F3-M2000	0.0049	0.0037	0.0035	24.7	0.776	0.342	0.356
F3-M4000	0.0027	0.0021	0.0020	25.3	0.835	0.350	0.361

Table S3: Scarce-reference scaling: ISE times the local-information rate index.

Scenario	M	p_M	Mp_Mh	X-anchor	Adaptive
SC-M1000-p0.025	1000	0.025	7.5	0.341	0.325
SC-M1000-p0.050	1000	0.050	15.0	0.347	0.296
SC-M2000-p0.025	2000	0.025	15.0	0.364	0.313
SC-M1000-p0.100	1000	0.100	30.0	0.344	0.286
SC-M2000-p0.050	2000	0.050	30.0	0.345	0.277
SC-M4000-p0.025	4000	0.025	30.0	0.348	0.283
SC-M1000-p0.200	1000	0.200	60.0	0.337	0.267
SC-M2000-p0.100	2000	0.100	60.0	0.329	0.263
SC-M4000-p0.050	4000	0.050	60.0	0.332	0.258
SC-M8000-p0.025	8000	0.025	60.0	0.338	0.263
SC-M2000-p0.200	2000	0.200	120.0	0.358	0.274
SC-M4000-p0.100	4000	0.100	120.0	0.332	0.256
SC-M8000-p0.050	8000	0.050	120.0	0.340	0.261
SC-M4000-p0.200	4000	0.200	240.0	0.319	0.241
SC-M8000-p0.100	8000	0.100	240.0	0.327	0.249
SC-M8000-p0.200	8000	0.200	480.0	0.323	0.250

Table S4: Raw-process multiplier-band coverage and width (target 0.95). Each cell uses 1,000 replications and 4,999 multiplier draws. Here $M = 2,000$ and $h = 0.30$, so $Mp_Mh = 30, 60, 120$; coverage Monte Carlo standard errors across both methods range from 0.0062 to 0.0091.

p	Ver.	Condition	Cov. Adaptive	Cov. anchor	Width Adaptive	Width anchor
0.05	R1	strong	0.934	0.927	0.390	0.511
0.05	R1	moderate	0.933	0.929	0.452	0.508
0.05	R1	weak	0.928	0.935	0.496	0.511
0.05	R1	locally failed	0.946	0.944	0.508	0.508
0.05	R2	strong	0.935	0.932	0.518	0.644
0.05	R2	moderate	0.910	0.908	0.582	0.633
0.05	R2	weak	0.926	0.921	0.620	0.632
0.05	R2	locally failed	0.917	0.919	0.643	0.641
0.10	R1	strong	0.948	0.942	0.275	0.362
0.10	R1	moderate	0.937	0.942	0.321	0.361
0.10	R1	weak	0.941	0.945	0.350	0.362
0.10	R1	locally failed	0.939	0.940	0.363	0.363
0.10	R2	strong	0.946	0.918	0.363	0.463
0.10	R2	moderate	0.939	0.921	0.416	0.460
0.10	R2	weak	0.946	0.939	0.444	0.456
0.10	R2	locally failed	0.940	0.938	0.458	0.458
0.20	R1	strong	0.945	0.932	0.201	0.256
0.20	R1	moderate	0.940	0.945	0.230	0.257
0.20	R1	weak	0.953	0.960	0.249	0.258
0.20	R1	locally failed	0.948	0.948	0.257	0.257
0.20	R2	strong	0.945	0.943	0.258	0.329
0.20	R2	moderate	0.936	0.938	0.295	0.328
0.20	R2	weak	0.948	0.944	0.314	0.324
0.20	R2	locally failed	0.945	0.946	0.327	0.327

Table S5: Raw-process coverage by the minimum number of verified kernel-support observations in any role fold, pooled over the 24 band cells.

Minimum role-fold count	Replications	Adaptive	Anchor
0–3	890	0.875	0.867
4–5	2,378	0.911	0.912
6–9	5,128	0.943	0.936
10+	15,604	0.945	0.943

Table S6: Vanishing-path components as the candidate-path scale s approaches zero. The final column is the mean Adaptive-minus-path-oracle ISE gap multiplied by 10^5 .

s	Relative curvature	$E(\widehat{\lambda}_h)$	$s E(\widehat{\lambda}_h)$	ISE gap ($\times 10^5$)
1.000	1.0000	0.8990	0.8990	6.010
0.500	0.2500	0.9986	0.4993	0.100
0.250	0.0625	0.9998	0.2499	0.016
0.125	0.0156	1.0000	0.1250	0.000
0.000	0.0000	0.0000	0.0000	0.000

16 Connections to adjacent methods

Table S7: Comparison with adjacent prediction-assisted, semi-supervised, validation-sample, and local-distribution methods.

Method	Primary target	Fixed continuous x_0	Selective MAR reference sampling	$p_M \rightarrow 0$	Entire CDF and simultaneous bands	Transfer or efficiency control
Validation-sample inference (Pepe, 1992; Tao et al., 2017)	Regression parameters under validation or two-phase measurement	No shrinking point target	Yes	Fixed-overlap theory	No local CDF result	Likelihood and semiparametric efficiency
PPI/PPI++ (Angelopoulos et al., 2023a,b)	Means and estimating-equation targets	No	Labeled and prediction samples	Not part of the main theory	Scalar or finite-dimensional inference	Prediction correction and scalar power tuning
Adaptive and safe semi-supervised inference (Song et al., 2024; Deng et al., 2024; Chen et al., 2025)	Finite-dimensional M-parameters or high-dimensional regression coefficients	No shrinking point target	Labeled and unlabeled, or selectively validated, samples	Not combined with local effective information	No local CDF process	Optimal, safe, or regularized auxiliary-information weights
PPCI (Sui et al., 2026)	Conditional functionals at a test point	Yes	Labeled and unlabeled samples in the main formulation	Not part of the main theory	Pointwise conditional-functional inference	Prediction-assisted localization and variance reduction
PAS (Li and Ignatiadis, 2025)	A collection of mean-estimation tasks	No	Gold-standard outcomes and predictions across tasks	Not part of the main theory	No conditional-CDF result	Empirical-Bayes shrinkage across tasks
Decaying-overlap semi-supervised inference (Testa et al., 2025)	General smooth semiparametric targets	No shrinking point target	Yes	Yes	Target-specific asymptotic normality	General influence-function theory under decaying overlap
Surrogate-assisted transported distribution inference (Wang, 2026)	Transported counterfactual CDFs and quantiles	No	Source validation subset and target covariates	No decaying-validation result	Uniform transported CDF and quantile bands	Efficient influence functions for transported distributional targets
Local distribution and conditional density (Cattaneo et al., 2024b,a)	Conditional distributions and densities	Yes	Fully observed outcomes	Not applicable	Yes	Local-polynomial distribution and density estimation
Present method	$F_0(\cdot x_0)$ and its kernel-localized version	Yes	Yes	Yes	Yes	Canonical gradient, exact transfer sign, common borrowing weight, and path-oracle equivalence

Convolution-smoothed prediction-powered quantile-regression estimators (Takeishi et al., 2026) target finite-dimensional coefficients with limited gold-standard outcomes, rather than a local CDF process under selective verification. The present method estimates a local conditional CDF when the reference outcome is selectively observed, a surrogate is broadly available, and the verification fraction may decrease. The canonical gradient gives the efficiency bound, and the quadratic criterion determines how much surrogate information to use. Sample splitting separates weight selection from CDF evaluation. The triangular-array limit then supports simultaneous CDF and quantile inference.

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